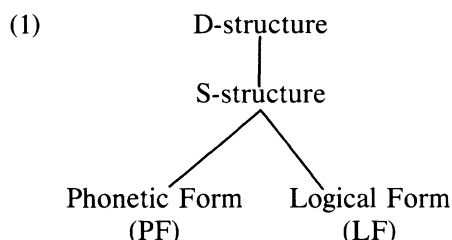


1. Properties of Logical Form

This article presents a proposal concerning the mapping of word structure onto the level of Logical Form (LF) in the grammatical model familiar from Chomsky and Lasnik (1977) and subsequent work:



Two minimal criteria of adequacy for LF representation seem to be the following:

- (2) a. All operators are found in operator ($\bar{A}$) positions, and their relative scopes are determined by the predicate “c-commands” (May (1977), Higginbotham (forthcoming)).
- b. The θ -Criterion is met: each θ -role is assigned to a grammatical function filled by an argument, each grammatical function filled by an argument bears a θ -role, and the relation between θ -roles and grammatical functions is bi-unique (Chomsky (1981)).

In this article I will argue for two somewhat less minimal empirical hypotheses about LF. Each has been advanced on the basis of evidence rather different from that presented here.

The first hypothesis is that of May (1977), which has been assumed in most sub-

This article is a revision of a paper written in the summer of 1983. It incorporates some relevant material from an earlier unpublished paper, Pesetsky (1979) (as indicated in the text). I am grateful to a number of people for comments on the earlier version, particularly Hagit Borer, Noam Chomsky, Mürvet Enç, Morris Halle, Larry Hyman, Osvaldo Jaeggli, David Lebeaux, Tom Roeper, Donca Steriade, Tim Stowell, Edwin Williams, and a very helpful anonymous LI reviewer. I am also grateful to the participants in morphology seminars at USC, the 1983 LSA Summer Institute at UCLA, and the University of Massachusetts at Amherst, and to the participants at the MIT Morphology Workshop and the Sloan Workshop on Lexical Phonology and Morphology at Stanford University, many of whom will find their influence on this revision.

sequent work on LF. May argued that the principal rule mapping S-structure representations onto LF is a rule that adjoins a category C to some node that dominates C. By this operation, which May called *QR* (for *Quantifier Rule*), an S-structure like (3a) is mapped onto an LF like (3b):

- (3) a. [_S John likes [_{NP} every girl]]
 b. [_S[_{NP} every girl]_i [_S John likes *e_i*]]

The second hypothesis about LF has been argued for in Pesetsky (1982, chap. 2) and in Pesetsky (forthcoming). There I proposed a somewhat new treatment of subcategorization, along the following lines. I argued that classical subcategorization frames conflate two types of information about lexical items: (a) what θ -roles they assign, and (b) what *types of syntactic categories* (NP, S', etc.) they select to receive their θ -roles.

An example will help to clarify this distinction. In an *Aspects*-style theory, the verb *receive* has a subcategorization frame of the following form:

- (4) receive: [+ ____ NP]

This subcategorization frame contains two pieces of information: that *receive* θ -marks an object, and that this object must be an NP. In Pesetsky (1982) I argued that the former information—about θ -marking proper—is the *only* information relevant to the *Projection Principle*: if a predicate θ -marks a constituent at some level L_j , then it θ -marks a constituent at all levels $L_{i...n}$. The latter information—about the “categorial selectional” properties of a predicate—is relevant *only at LF*, and crucially is not relevant at other levels like D-structure and S-structure.

In practical terms, this means that a constituent can change its categorial specification in the course of a derivation, so long as the “correct” category is present by LF. Consider again the verb *receive*. This verb θ -marks an object; hence, by the Projection Principle, *some* object must be present at D-structure, at S-structure, and at LF. But what kind of object? As the traditional subcategorization frame states, *receive* “categorially selects” (*c-selects*) an NP. By the new hypothesis in Pesetsky (1982), although some object or other is present at all levels, this object is *required* to be an NP only at LF. At D-structure and at S-structure, it may bear completely different categorial features.

In Pesetsky (1982) I argued that examples of exactly the expected type are found in Russian and explain some puzzling subject-object asymmetries in that language. In particular, I argued that certain phrases in NP positions in Russian (some types of numeral phrases, “genitives of negation,” etc.) have quantifier heads, and are not NPs, but QPs. Thus, for example, representations like (5) are acceptable in Russian at S-structure:

- (5) *S-structure*
 [_S ja polučal *QP*] ‘I received QP’

This representation is fine at S-structure, since c-selection does not apply there. How does (5) satisfy c-selection at LF? I claimed that QR can adjoin the QP to its S. The Projection Principle requires that some kind of empty “trace” be left behind to bear the object θ -role, but it does not specify what category that trace must belong to. Why not the NP that c-selection requires? C-selection can thus be satisfied at LF, with the following representation:

(6) *Logical Form*

$[_S QP_i [_S \text{ja polučal } [_{NP} e]_i]]$

The analysis is motivated by the interaction of structures like (6) with Chomsky’s (1981) *Empty Category Principle (ECP)*. The object NP trace of QP in (6) satisfies the ECP because it is properly governed by the verb *polučat* ‘receive’. But suppose such an NP trace bound by a QP were found in subject position. It could only satisfy the ECP if it were properly bound by its coindexed antecedent. Here the categorial mismatch between NP trace and QP antecedent might well be important. An LF structure like (7) would quite plausibly violate the ECP, and indeed subject QPs (the numeral and genitive phrases mentioned above) are quite regularly excluded in Russian:

(7) *Logical Form*

$*[_S QP_i [[_{NP} e]_i \text{Infl polučal NP-object}]]$

I have done my best to sketch some motivation for the hypothesis that c-selection applies at LF only, without giving full details of the Russian argument. Assuming that the hypothesis is clear, I refer the reader to Pesetsky (1982; forthcoming) for the necessary further details.

Here I argue that some very different facts—facts about *word structure*—support both above-mentioned hypotheses about LF, namely:

- (8) a. LF is derived from S-structure by adjunction of a category C to a node that dominates C. (May (1977))
- b. Categorial selection applies only at LF. (Pesetsky (1982; forthcoming))

Word structure also supports the most general property of LF, as proposed by Chomsky (1976): that LF is the immediate linguistic input to semantic interpretation.

The domain of investigation will be “bracketing paradoxes” in morphology.

2. Bracketing Paradoxes

Williams (1981b), Pesetsky (1979), Kiparsky (1983), and others have presented evidence suggesting that the structure of words and their interpretation are not always characterizable by a unique labeled tree. For the most part, the observations have been treated as “paradoxes” to be handled by some mechanism peculiar to morphology. In this section I consider three such “paradoxes” and suggest a general explanation for these phenomena.

2.1. -er

I begin with some facts introduced in Pesetsky (1979), which concern the interaction of the comparative suffix *-er* with the negative prefix *un-*.

As is known, the comparative suffix *-er* is subject to an odd phonological condition. Roughly speaking, it may attach to monosyllabic adjectives and to a limited class of possibly disyllabic adjectives with a very light second syllable (a syllabic sonorant or the glide [y]):

- (9) a. blacker, softer, riper, tougher, truer, poorer, swifter, nicer, . . .
 b. happier, heavier, luckier, kindlier, rottener, subtler, abler, pleasanter, ?chipperer, . . .

Crucially, it is difficult or impossible to attach comparative *-er* to disyllables with a heavier second syllable or to any word with three or more syllables:

- (10) a. *directer, *activer, *complexer, *Klamather, *porouser, . . .
 b. *eloquenter, *irascibler, *Etruscaner, *importanter, *loquaciouser, . . .

Now, however, some apparent counterexamples arise:

- (11) unhappier, unpleasanter, unluckier, . . .

At first sight, *-er* appears to be attaching to the trisyllabic bases *unhappy*, *unpleasant*, *unlucky*, in violation of the previous generalization. It is not possible to claim that the addition of *un-* completely nullifies the phonological generalization about *-er*, since addition of *un-* to a word that does not allow *-er* in the first place, like those in (10), does not improve matters:

- (12) *uncomplexer, *unporouser, *uneloquenter, *unirascibler, *unloquaciouser, *unimportanter, . . .

As a solution, we may easily assume that the prefix *un-* is attached by lexical insertion or word formation rule *after* attachment of the suffix *-er*. Thus, when *-er* is added to the first word in (11), it is being added to [*happy*], and not to [*un* [*happy*]]. The phonological condition on *-er* attachment is not violated, and we correctly predict that *un* + A + *er* will not violate this phonological condition iff A + *er* does not.

The phonological condition on *-er* thus motivates the following structure:

- (13) [_A un [_A happy er]]

As noted in Pesetsky (1979), this solution to the phonological problem immediately raises a problem of interpretation. In the structure shown in (13), comparative *-er* is in the scope of negative *un-*, but this does not at all correspond to the interpretation of the word. *Unhappier* does not mean 'not more happy', as structure (13) implies. Rather, it means 'more not happy', as in structure (14):

- (14) [_A [_A un happy] er]

In (14) *un-* and *happy* are sisters, completely spoiling the phonological generalization that *-er*'s sister cannot be a trisyllable, but correctly capturing the scope relations in the word.

The "paradox" thus takes the following form:

(15)	[Prefix [root + suffix]]	[[Prefix + root] suffix]
	Satisfies phonological conditions	Need not satisfy phonological conditions
	Logical scope not given by c-command	Logical scope given by c-command

I will argue that (15) is not a paradox at all. Rather, it presents a classic argument for two levels of representation. Phonological constraints apply to the level on the left. Logical scope is indicated at the level on the right. Before speculating further, let us consider some other bracketing problems of a similar nature.

2.2. Russian Yer Lowering

This second paradox, which was brought to my attention by Morris Halle, is also raised in Pesetsky (1979).¹ Russian and other Slavic languages exhibit a well-known group of vowel~zero alternations (see among others Lightner (1972), Gussmann (1980)), exemplified here in Russian:

(16) a.	nemec	(nom sing)	nemc-a	(gen sing)	'German'
	jasen	(masc sing)	jasn-o	(neut sing)	'clear'
b.	okon	(gen plur)	okn-o	(nom sing)	'window'
	rot	(nom sing)	rt-a	(gen sing)	'mouth'

I will briefly sketch the standard analysis of Lightner and others. These [e]~zero and [o]~zero alternations are due to a rule of vowel deletion rather than insertion, given minimal or near-minimal pairs like the following:²

(17) a.	interesen	(masc sing)	interesno	(neut sing)	'interesting'
	son	(masc sing)	sna	(gen sing)	'dream'
b.	lasok	(gen plur)	laska	(nom sing)	'weasel'
	lask	(gen plur)	laska	(nom sing)	'caress'

Following historical and generative tradition, I call the underlying vowels *yers*. Other alternations in Russian suggest that the underlying vowel responsible for [e]~zero alternations is a lax *ɪ*, and that the vowel responsible for [o]~zero alternations is a lax *ʊ*.

¹ See Booij and Rubach (1983) for the same paradox and a similar solution in Polish.

² (17a) shows that an insertion rule could not predict the nature of the inserted vowel; (17b) shows that we cannot even predict whether a vowel is inserted at all.

Plainly, two rules are needed: one lowers *ɪ* and *ʊ* to *e* and *o* in certain environments; the other deletes all yers that have not been lowered by the first rule:

(18) a. *Yer Lowering*

$\text{ɪ, ʊ} \rightarrow \text{e, o} / \text{some environment}$

b. *Yer Deletion*

$\text{ɪ, ʊ} \rightarrow \emptyset$

Yers lower in two obvious environments: in final syllables and before certain, but not all, suffixes:

- | | | | |
|---------|--------------------------------|------------------|----------------|
| (19) a. | den' (nom sing) | dn'-a (gen sing) | 'day' |
| b. | den'-k-a (gen sing) | | <dimin.> |
| | but dn-ev-n-oj (masc nom sing) | | 'day' (adj) |
| (20) a. | len (nom sing) | l'na (gen sing) | 'flax' |
| b. | len-k-a (gen sing) | | <dimin.> |
| | but l'n-jan-oj (masc nom sing) | | 'flaxen' |
| (21) a. | okon (gen plur) | okno (nom sing) | 'window' |
| b. | okon-n-oj (masc nom sing) | | 'window' (adj) |

Thus, Yer Lowering can be rephrased as follows:

(22) *Yer Lowering*

$\text{ɪ, ʊ} \rightarrow \text{e, o} / \text{--- } C_0 \left\{ \begin{array}{l} \#\# \\ \text{certain suffixes} \end{array} \right\}$

But these "certain suffixes" themselves contain yers:

- | | | |
|----------|-----------------------|-----------------|
| (19) b'. | den'-ok (nom sing) | 'day' <dimin.> |
| | (cf. (19b) den'-k-a) | |
| (20) b'. | len-ok (nom sing) | 'flax' <dimin.> |
| | (cf. (20b) len-k-a) | |
| (21) b'. | okon-en (masc sing) | 'window' <adj.> |
| | (cf. (21b) okon-n-oj) | |

Therefore, we can revise Yer Lowering as follows:

(23) *Yer Lowering*³

$\text{ɪ, ʊ} \rightarrow \text{e, o} / \text{--- } C_0 \left\{ \begin{array}{l} \#\# \\ \text{ɪ} \\ \text{ʊ} \end{array} \right\}$

Recall now that the yer *deletion* rule that follows (23) deletes all unlowered yers.

³ For maximum comprehensibility, I have left (23) in somewhat sloppy form. *ɪ*, *ʊ* lower to [e] and [o], respectively, but the trigger in the next syllable may be *ɪ* or *ʊ* regardless of the type of yer that is lowering. I leave unexplored the obviously interesting possibility of exploiting modern nonlinear phonology in a re-statement of (23).

This allows us to eliminate “##” from the environment of (23), by simply giving each Russian word that ends in a consonant on the surface an underlying final yer. This yer will trigger lowering in the preceding syllable and will subsequently disappear.

This much is the standard analysis. We are left with the ordered rules in (24):

(24) a. *Yer Lowering*

$$i, u \rightarrow e, o / \text{ — } C_0 \begin{Bmatrix} i \\ u \end{Bmatrix}$$

b. *Yer Deletion*

$$i, u \rightarrow \emptyset$$

Sample derivations are as follows:

(25)	a.	b.	c.
	'day' (nom sing)	(gen sing)	(gen sing) <dimin.>
	din'-i ⁴	din'-a	din'-ik-a
Lowering	den'-i	—	den'-ik-a
Deletion	den'-	dn'-a	den'-k-a

Now, what happens when a number of yers occur in successive syllables? Consider first a noun: the double diminutive of the word for 'day'. To the root *din'*, we can add two occurrences of the diminutive suffix *-ik* and the nominative desinence *-u*. The correct output lowers all yers but the last. We can achieve this if Yer Lowering applies iteratively from left to right:

(26)	din'-ik-ik-u	'day' <double dimin.>
Lowering	e	
	e	
	e	
	—	
Deletion	∅	
Other rules	den'-oč-ek	

But consider now a prefixed verb: in particular, the prefix *podu-* followed by the root *žig* 'burn', past tense suffix *-l-*, and masculine desinence *-u*. Here, left-to-right application yields an incorrect result:

(27)	podu-žig-l-u	'set on fire' (past masc)
Lowering	o	
	e	
	—	
Deletion	∅	
Other rules	* <i>podo-žog</i>	

⁴ The analysis or status of palatal consonants like the [n'] in forms of the word 'day' is irrelevant here.

Right-to-left application, however, yields exactly the correct result:⁵

- (28) *podu-žig-l-u*
 Lowering —
 e
 —
 Deletion $\emptyset$ $\emptyset$
 Other rules *pod-žog*

Right-to-left application of Yer Lowering to the double diminutive noun, however, yields the wrong result. Compare the following derivation with (26):

- (29) *din'-ik-ik-u*
 Lowering —
 e
 —
 e
 Deletion $\emptyset$ $\emptyset$
 Other rules **den'-čok*

Surely Yer Lowering does not apply in one direction in nouns and in another direction in verbs. As a solution, suppose that the direction of iteration actually mirrors the structure of the words undergoing the rule—in other words, that Yer Lowering is cyclic. In fact, the left-to-right order in the double diminutive noun precisely mirrors the natural cyclic order. To make a cyclic derivation work, we must assume that Yer Deletion is postcyclic:⁶

- (30) [[[[[*din'*] *ik*] *ik*] *u*]
 Lowering
 cycle 1 —
 cycle 2 e
 cycle 3 e
 cycle 4 e
 cycle 5 —
 Deletion $\emptyset$
 Other rules *den'-oč-ek*

⁵ In some forms of the verb—for example, the feminine—the direction makes no difference:

- (i) *podu-žig-l-a* *podu-žig-l-a*
 Lowering o —
 — o
 Deletion $\emptyset$ $\emptyset$
 Output *podo-žg-l-a*

⁶ In Pesetsky (1979) I argued that Yer Lowering has, besides cyclicity, other properties that identify it as a lexical rule. In this it contrasts with Yer Deletion, which is postlexical.

A cyclic order also derives the prefixed verb correctly, but we must assume a rather interesting structure for the verb; it is here that the bracketing “paradox” arises. If we assume that the root and suffixes form a constituent and that the prefix constitutes the *last* cycle, cyclic Yer Lowering works correctly:

(31)	[podu [[[žig] l] u]]	
Lowering		
cycle 1		—
cycle 2		—
cycle 3		e
cycle 4	—	
Deletion	∅	∅
Other rules	<i>pod-žog</i>	

This structure, with the prefix outermost, allows us to have one Yer Lowering rule, applying cyclically, with no bizarre stipulations about the direction of iteration. So far so good, but there is a problem. The structure seen in (31) is necessary if the phonology is to function smoothly, but it is quite odd from the point of view of the semantics. In particular, consider how the individual morphemes in a word like *podžog* contribute to the meaning of the word.

In Russian, as in many languages, the combination of a verbal root and a prefix often yields an unpredictable, idiosyncratic meaning. In the present case, the verbal root *žig*, by itself, means ‘burn’. The prefix *podu-* often seems to mean something like ‘up to’. The combination of this prefix with the root *žig*, however, yields the meaning ‘ignite’ ‘set fire to’—not such a wildly drifted meaning, but still unpredictable. On the other hand, the past tense suffix *-l* and the masculine desinence *-u* make thoroughly predictable contributions to the interpretation of *podžog* and similar verbs.

A natural principle governing the assignment of structures to words is the *compositionality* requirement: the requirement that semantic units (like the *prefix* + *verb* under discussion) form morphological constituents. I will offer empirical support for this principle later, in section 3. At the moment, however, I will simply note that the structure the phonology assigns to *podžog* in (31) is clearly at variance with it. Thus, *podžog* exhibits a conflict between phonologically motivated bracketing and semantically motivated bracketing, of a type now familiar from the earlier discussion of *-er*. Harking back to the table in (15), we may represent the present conflict as follows:

(32)	[Prefix [root + suffix]]	[[Prefix + root] suffix]
	Undergoes the phonology	Does not undergo the phonology
	Does not satisfy a semantic compositionality requirement	Satisfies semantic compositionality

2.3. *un* # grammatical + *ity*

The bracketing problems just discussed arise from conflicts between the phonology and interpretation. The third “paradox” arises from a conflict between the subcategorization frame for the affix *un-* and the Level Ordering Hypothesis of Siegel (1974) and Allen (1978). This problem is well known from the literature (see, for example, Williams (1981a)), so my presentation of it will be brief.

It is a commonplace that an affix often specifies the category of its sister node and the category of the node that immediately dominates it after affixation. Thus, English *-ness* attaches only to adjectives and is immediately dominated by N. Following Lieber (1980) and Selkirk (1982), let us assume that specifications of this sort belong in the lexical entries for affixes. In particular, let us assume with Lieber and Selkirk that the specification of an affix’s sister node is given in a *subcategorization frame* for that affix. As I discuss later, I assume with Lieber that percolation conventions will generally project the *categorial specification* of an affix to the node immediately dominating that affix (with certain exceptions). Thus, the lexical entry for the affix *-ness* will look something like (33):

- (33) *ness*: N, [+ A ____]

Siegel (1974) has shown that the negative prefix *un-* must be subcategorized to attach only to adjectives. Apparent cases of negative *un-* in nouns actually result from prior attachment to an adjectival base. In other words, *un-*’s sister is always A:

- (34) a. [[*un* [_A happy]] *ness*]
 b. [[*un* [_A true]] *th*]
 c. *[*un* [_N cheer]]

The subcategorization frame for *un-* is therefore [+ ____ A]. As for the categorial specification of *un-*, there are two a priori possibilities. (This issue will be important later.) It seems to be a fact (ignoring the upcoming “paradox”) that when *un-* attaches to an adjective, the result is an adjective as well. Thus, we might stipulate that *un-* belongs to category A.

Work done by Lieber (1980, 50) raises another possibility, however. She studies the prefix *counter-*, which, she notes, may attach freely to the categories N, V, and A (*counterespionage*, *countersign*, *counterintuitive*). The node that dominates *counter-* inherits its category, not from *counter-*, but from *counter-*’s sister:

- (35) a. [_N *counter* [_N espionage]]
 b. [_V *counter* [_V sign]]
 c. [_A *counter* [_A intuitive]]

Affixes like *counter-*, which project the category of their sister, Lieber assigns to a “null

category class'' denoted by the symbol $\emptyset$. Since *un-* takes an A as its sister and is dominated by A, we might say that *un-*, like *counter-*, belongs to the null category class and has the lexical entry in (36). This will be our assumption, for reasons that will shortly be clear:

(36) *un*: $\emptyset$, [+ ____ A]

We now turn to another property of affixes, which is relevant to the paradox presented in this section. Siegel (1974) observed an important regularity in the nesting of English affixes, which she and Allen (1978) developed into the *Level Ordering Hypothesis*. Roughly, it appears that English affixes belong to one of at least two classes, or Levels.⁷ Affixes belonging to Level I (demarcated, for Siegel and Allen, by a “+” boundary symbol) are attached earlier than, and hence are c-commanded by, all affixes belonging to Level II (demarcated by “#”). Thus, although the affix *-less* creates adjectives, and although the affix *-ity* is subcategorized to attach to adjectives, a word like **hopelessness* is ruled out by Level Ordering:

(37) *[[[hope_N] #less_A] +ity_N]

The suffix *-ity* is Level I, and *-less* is Level II. Since *-ity* is attached after *-less* and c-commands it, the Level Ordering Hypothesis is violated.

By contrast, the word *analyzability* satisfies the Level Ordering Hypothesis, since the Level I affix *-ity* c-commands only other Level I affixes (here, *-able*):⁸

(38) [[[analyze_V] +able_A] +ity_N]

Finally, *?analyzableness*, though perhaps “blocked” (in the sense of Aronoff (1976)) by the existence of (38), is also grammatical, since no Level I affix c-commands any Level II affix:

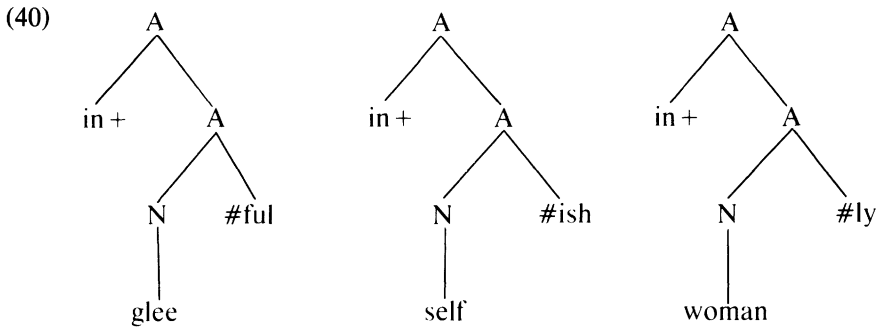
(39) [[[analyze_V] +able_A] #ness_N]

Allen (1978) has argued that the negative prefix *un-* belongs to Level II. Many of the arguments involve distinguishing the properties of *un-* from the properties of the negative prefix *in-*. For example, whereas both *un-* and *in-* generally attach to adjectives (with some exceptions for *in-*, which Allen notes), *in-* does not generally attach to adjectives created by level II affixation: **ingleeful*, **inselfish*, **inwomanly*, etc. This can

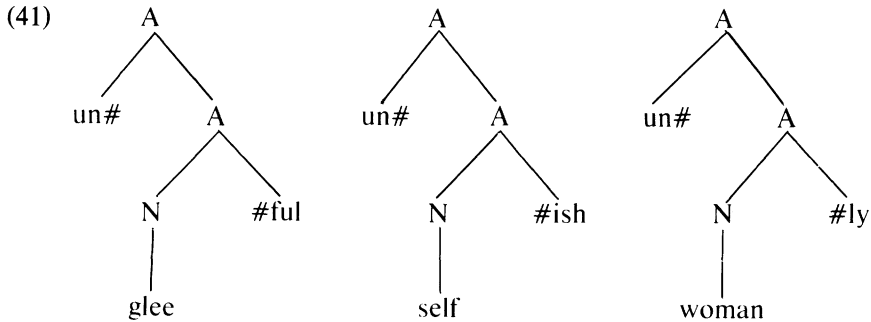
⁷ Capitalized “Level” refers to “Levels” in Siegel’s and Allen’s sense. I use lower-case “level” elsewhere, e.g. “level of representation.”

⁸ Following Kiparsky (1983), I assume that the *-able* in forms like (38) is indeed Level I, contra Aronoff (1976). Kiparsky argues that its apparent Level II behavior is due to its being marked extrametrical.

be explained if *in-* is Level I:

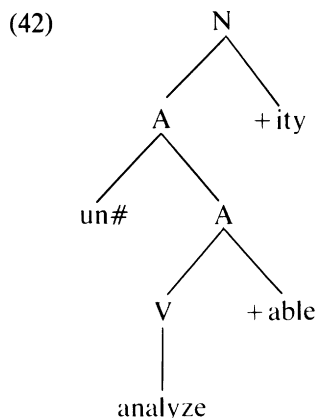


On the other hand, *un-* is acceptable in place of *-in* in each case of (40), suggesting that *un-* is Level II:



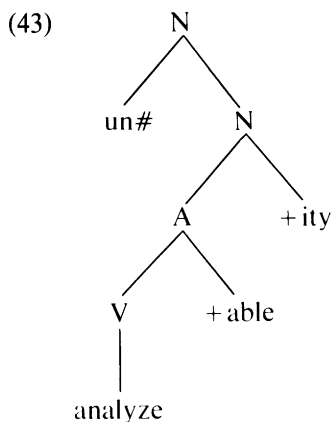
Other evidence concerns stress (*unfinite* vs. *infinite*) and the application of nasal assimilation (*u[n]lawful* vs. *i[l]legal*).

However, as has been noted since Siegel (1974), a “paradox” arises if *un-* is a Level II affix that subcategorizes adjectives. The paradox arises because of the grammaticality of words like *ungrammaticality* or *unanalyzability*, if we borrow from syntax the standard assumption that elements subcategorize for sister nodes. (This assumption will be empirically supported below in the discussion of (69).) Subcategorization forces the structure in (42), where *un-* is a sister of the adjective *analyzable*:



Structure (42) satisfies subcategorization, but should violate the Level Ordering Hypothesis, since the Level I suffix *-ity* c-commands the Level II prefix *un-*.

The Level Ordering Hypothesis forces a structure like (43) instead:



This structure violates the lexical requirements of *un-*. The fact that it is immediately dominated by N can be explained easily if *un-* belongs to the “null category class” $\emptyset$, as suggested in (36). The fact that its sister is not an adjective, however, is not so easily explained. Once more, as with *unhappier* and Russian *podžog*, we have a conflict of requirements that no unique structure can satisfy:

(44)	[Prefix [root + suffix]]	[[Prefix + root] suffix]
	Satisfies Level Ordering Hypothesis	May violate Level Ordering Hypothesis
	May violate subcategorization	Satisfies subcategorization

2.4. Logical Form

The claim that structure (43) “violates subcategorization” is actually a bit too strong. Subcategorization frames for affixes, like subcategorization frames for words, indicate two things. Consider again a frame for *un-* like [+ ____ A]. This frame indicates, first, that *un-* must be a sister to something, and a left-hand sister. This is somewhat like the information that *receive* (or its Russian equivalent) assigns a θ -role to an object, which is therefore obligatorily present. The frame also indicates that this sister must belong to a particular category—in this case, A. It could be that in morphology, as in syntax (if Pesetsky (1982) is correct), these two sorts of requirements, the “positional” and the “categorial,” should not be conflated into a single subcategorization frame. Both structures for *unanalyzability*, (42) and (43), satisfy the “positional” requirement that *un-* be a left-hand sister to something. What (43) violates, and (42) does not, is the “categorial” requirement that *un-* be a sister to an *adjective*. A more precise statement of the paradox would be (45):

(45) [Prefix [root + suffix]]	[[Prefix + root] suffix]
Satisfies Level Ordering Hypothesis	May violate Level Ordering Hypothesis
May violate c-selection	Satisfies c-selection
Satisfies positional requirements of subcategorization	Also satisfies positional requirements of subcategorization

Let us now make the following hypothesis: the three “paradoxes” discussed above are instances of the same phenomenon.⁹ Looking at the three together can point the way to a solution that could not be discovered by looking at any one of the three by itself.

I thus suggest that words like *unhappier*, *podžog*, and *ungrammaticality* have two distinct structures, which obtain at different levels of representation, much as sentences may have distinct representations at D-structure, S-structure, LF, or other levels. We can see the properties of these levels if we put together all the facts we now know about them:

(46) *Level of Representation 1*

Prefix c-commands root and suffixes.

Phonological restrictions on affixation are satisfied.

⁹ Indeed, phenomena of this sort appear to be widespread. Ha (1984) gives an argument from Korean that is quite similar to my argument from Russian, though the phonological rule involved is of course different. Kitagawa (1984) presents a bracketing paradox in Japanese, involving the interaction of the *rendaku* rule with otherwise motivated syntactic structures for compounds.

Phonological rules apply to this level's structures.
 Logical scope relations are *not* necessarily given by c-command.
 Does *not* necessarily satisfy semantic compositionality.
 C-selection is *not* necessarily satisfied.
 "Positional" (noncategorical) aspects of subcategorization are satisfied.

Level of Representation 2

Suffixes c-command root and prefix.
 Phonological restrictions on affixation are *not* necessarily satisfied.
 Phonological rules do not apply (apparently) to this level's structures.
 Logical scope relations are given by c-command.
 Satisfies semantic compositionality.
 C-selection is satisfied.
 "Positional" (noncategorical) aspects of subcategorization are satisfied.

In section 1 I suggested that three properties characterize the syntactic level of Logical Form: (a) scope relations among operators are determined; (b) the "positional" requirements of the θ -Criterion are satisfied; (c) c-selection applies. I also claimed that at other levels of representation (D- and S-structure), (a) scope relations are *not* necessarily determined by the structure; (b) positional requirements are also met, because of the Projection Principle; (c) c-selection does *not* apply.

The conclusion should be clear. "Level of Representation 2" meets all the criteria for LF, and "Level of Representation 1" meets those for a non-LF level of representation. We may call level 1 "S-structure," since it is clearly the input to the phonology. We come to grips with our morphological "paradoxes" if we assume that the constituent structure of words at S-structure is not necessarily isomorphic with their constituent structure at LF.

The picture that emerges is esthetically pleasing. Word structure at S-structure is determined by purely "morphological" information. Quirky phonological conditions on affixation obtain at S-structure, as do language-specific constraints like Level Ordering. Phonological rules like Yer Lowering apply to the bracketed representations created at this level. As expected, c-selectional requirements of affixes (e.g. "my sister node is an adjective") do not have to be met, nor is logical scope directly determined by S-structure representations.

On the other hand, word structure at LF is determined by more "semantic" information. C-selection—plausibly related to the semantics of various affixes—must be satisfied. C-command determines logical scope. Semantic units are structurally represented. Purely morphological considerations like Level Ordering or phonological shape are irrelevant.

Our proposal is thus satisfying from the point of view of "modularity," and it seems in accord with what has been suggested elsewhere about LF and other levels of representation. The next task is to determine the rule or rules that map morphological

S-structure representations onto LF. Concretely, how do prefixes like *un-* or Russian *podu-*, which c-command root and suffixes at S-structure, come to be c-commanded by the suffixes at LF? Most of the interesting work will be done by the rule that is already motivated to map syntactic S-structure onto LF representations: May's rule of QR. Before discussing this in detail, however, I will present an argument that these "paradoxes" really do provide evidence for distinct levels of representation.

3. Levels of Representation and Lexical Idioms

In this section I add another property to the level of LF for words and in so doing argue explicitly that words may have distinct representations at more than one level. The discussion concerns "lexical idioms."

First consider how word structure is determined at S-structure (an examination that will mostly follow Lieber (1980)). An affix like English *-al* has the following lexical entry:

- (47) *al*: A, [+N ____]

The subcategorization frame for *-al* indicates that it must be the unique sister of a constituent of category N. We thus know that the structure in (48) is well-formed:

- (48) [[*industry*_N] *al*_A]

Consider now an affix like *-ize*:

- (49) *ize*: V, [+A ____]

(49) indicates that the following structure is also well-formed:

- (50) [[[*industry*_N] *al*_A] *ize*_V]

The idea that lexical properties, interacting with independently given principles, determine the constituent structure of words is exactly parallel to the approach to phrase structure developed by Chomsky (1981; 1982) and Stowell (1981) and does not differ significantly from Lieber (1980). Also, the fact that *-al* and *-ize* are "bound morphemes" simply follows from the fact that they have subcategorization frames. Furthermore, if we suppose, as suggested in Pesetsky (1979) and subsequently, that affixation in the lexicon applies in ordered blocks, we gain the results of the Level Ordering Hypothesis. In particular, we need not assume that word structure is determined by a phrase structure grammar, as suggested in Selkirk (1982).

To determine the category of the node immediately dominating morphologically complex material, we adopt the essentials of Lieber's (1980) Percolation Conventions:

- (51) *Percolation Conventions*

Convention I: All features of a stem morpheme, including category features, percolate to the first nonbranching node dominating that morpheme. [Hence the *N* dominating *industry* in (48) and (50).]

Convention II: All features of an affix morpheme, including category features, percolate to the first branching node dominating that morpheme. [Hence the *A* dominating *-al* in (48) and (50).]

Convention III: If a branching node fails to obtain features by Conventions I or II, features from the next lowest labeled node are automatically percolated up to the unlabeled branching node. [This covers transparent, $\emptyset$ -category affixes like *counter-* and, I claim, *un-*.]

We see the action of these conventions in structures like (50). The node immediately dominating *industry* is labeled by Convention I, and the nodes immediately dominating *-al* and *-ize* are labeled by Convention II.

Let us now look at Convention II more closely. Selkirk (1982) raises a problem with respect to this convention that we will solve by an appeal to properties of LF. This appeal, in turn, will lead to the argument for the independence of LF representations for words.

Percolation Convention II states that affixal features percolate to the immediately dominating branching node. Williams (1981a) calls such affixes the “heads” of their constituents. Selkirk (1982, 98ff.) observes that, in syntax, heads of constituents not only percolate their features to dominating nodes, but also are uniquely granted the power to subcategorize their sister nodes. Thus, in the VP, verbs subcategorize their objects, but objects do not subcategorize their verbs. Selkirk suggests that such considerations may well carry over to morphology.

In a word like *analyzable*, *-able* has two of the properties of a phrasal head in syntax: it percolates its features and subcategorizes its sister node. The morpheme *analyze* in this word, on the other hand, does not percolate its features up, nor does it subcategorize for its sister *-able*. *Analyze* is to *-able* in *analyzable* as an NP object is to the verb in a VP. This observation in turn raises the possibility of explaining Convention II in terms of more general properties of heads and nonheads.

Selkirk does not wholly adopt this explanation, however, because of the existence of bound nonaffixes like *ceive* in *deceive*. In *deceive*, both *de-* and *ceive* are bound. If “boundness” is always due to a subcategorization frame, and if only heads have subcategorization frames, then *deceive* illegally has two heads. If we wish to derive Convention II from more general properties of heads, and if we maintain that only heads subcategorize, we must find another account of “bound roots” like *ceive*.

Following Selkirk’s lead, we might say that *ceive* is a bound morpheme because it possesses a subcategorization frame, e.g. [+ Affix ____]. This does not solve the problem, since it does not explain the contrasts in (52):

- (52) a. deceive, receive, perceive, . . .
 b. *transceive, *proceive, *inceive, *interceive, . . .

Let us approach the problem of words like **transceive* or **proceive* by making two rather naive observations about the behavior of bound morphemes like *ceive*. First, the meaning of words like *deceive* and *receive* is not compositionally derived from the meanings of its component morphemes in any obvious way. (The same issue arose in the discussion of Russian prefixed verbs.) Second, what is wrong with a word like **transceive* or **proceive* is rather simple: *it doesn't mean anything*. In this, our intuitions contrast with our intuitions about, say, **ceivetrans* or **analyzitable* (on which, see section 4.3), which seem to represent unlawful concatenations in the first place.

In the normal case, the meaning of a morphologically complex word is a straightforward function of the meanings of its constituents. In some cases, however, a constituent may have no meaning associated with it. Instead, a meaning is arbitrarily associated with the combination of that constituent with some other. Such is the case, for example, with *deceive*. The morpheme *ceive* by itself has no meaning, nor does this occurrence of *de-*.¹⁰ The combination of the two morphemes does, however, have a meaning associated with it. Let us call the idiosyncratic rules that interpret pairs like *de- + ceive* “rules of idiosyncratic interpretation,” as in (53):

(53) *de- + ceive* → ‘successfully lie to’

We may say that a morpheme is *interpreted* iff it is lexically associated with a meaning or participates in a rule of idiosyncratic interpretation. We may now place a star on words like **proceive*, **transceive*, or, for that matter, **ceive*, by means of the following filter:

(54) **W*, where *W* is a word containing an uninterpreted morpheme.

We rule out **ceive* and **proceive* because no morpheme in these words is interpreted. Other, slightly more complex cases also follow from (54). In a word like *transact*, *act* has a meaning lexically associated with it in any case, but a more specific rule of idiosyncratic interpretation assigns a meaning to *act* in combination with *trans-*.

Clearly, if we invoke rules of idiosyncratic interpretation and a filter excluding words with uninterpreted morphemes, we have an account for **proceive*, **ceive*, etc., that does not rely on subcategorization frames. We can now derive Convention II from general properties of heads, just as Selkirk suggests. With this result in hand, let us proceed to another parallel between word-internal rules and sentence-level rules.

I have proposed that rules at or past LF may assign idiosyncratic interpretations to particular combinations of morphemes. Such rules are not unknown at the sentence level, where the combinations of words to which they apply are known as *idioms*. At the sentence level, as word-internally, rules of idiosyncratic interpretation may assign

¹⁰ Reduction of the [e] in *d[e]ceive* shows that the *de-* here is a Level I prefix distinct from the Level II *de-* in *delouse*. The latter *-de* does have a meaning, as a “reversative” prefix.

interpretations to formatives that have no meaning in isolation. For example:

- (55) a. John left me in the *lurch*.
(‘left me abandoned and vulnerable to attack’)
- b. He protects his *kith* and kin.
(‘close relatives’)
- c. Mary played *hooky*.
(‘failed to attend school’)
- d. We *loused* it up.
(‘made a mess of it’)
- e. You *grossed* me out.
(‘thoroughly disgusted me’)

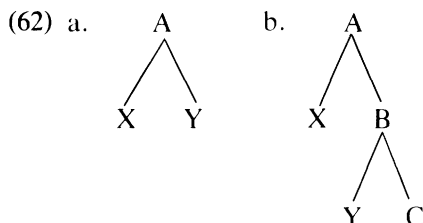
These idioms are very much like *transact*. Although some elements of the idioms (*left*, *up*, *kin*, *played*, etc.) have lexically assigned meanings, the italicized elements do not—at least in most speakers’ vocabularies. Nonetheless, rules of idiosyncratic interpretation apply to the combinations of words above and provide them with interpretations. Furthermore, in syntax and at the word level, a rule of idiosyncratic interpretation may apply optionally to preempt a normal compositional reading for a phrase. In syntax, such idioms are rather common:

- (56) a. Bill kicked the bucket.
(‘died’)
- b. You’ll eat your words.
(‘regret what you said’)
- c. Let’s talk turkey.
(‘be frank’)
- d. Mary gave John the finger.
(‘directed an obscene gesture at John’)

Such idioms exist word-internally as well. For example:

- (57) a. We were shocked at the *rarity* of this book.
- b. This old tome is a real bibliographic *rarity*.
- (58) a. John’s *peculiarity* is matched only by his brilliance.
- b. A *peculiarity* of the IBM keyboard is the odd location of its shift key.
- (59) a. Haig’s *recession* into obscurity was swift.
- b. The current *recession* put millions out of work.
- (60) a. The *installation* of the elevator took a week.
- b. Smoking is prohibited on the grounds of the nuclear *installation*.
- (61) a. Your *representation* of Bill’s views is correct.
- b. Consider the LF *representation* for the Moldavian sentence above.

In the discussion of Russian prefixed verbs in section 2.2 I suggested that “semantic units” defined by such rules should “form morphological constituents.” For example, one might expect morphemes X and Y to undergo a rule of idiosyncratic interpretation in (62a), but not in (62b):



Such a locality restriction strongly recalls familiar restrictions on sentence-level idioms (Chomsky (1965), Marantz (1984)). Suppose that both phrasally and word-internally *for idioms of two members X and Y, X and Y must be sisters*.¹¹ At the sentence level, this explains the absence (or paucity) of verb-subject idioms. What results obtain word-internally?

This locality restriction was essential inside the word if the Russian prefixed verbs were to yield an interesting “paradox.” One might reasonably respond to that earlier discussion with the suggestion that the paradox be solved by abandoning the locality requirement on idiosyncratic interpretation. I now give an independent argument for the locality restriction and, with it, for the existence of LF for words.

Consider a word like *rarity* in (57). (57a) shows a compositional reading for this word: ‘the fact that X is rare’ or ‘the degree to which X is rare’. (57b) shows a reading that depends on a rule of idiosyncratic interpretation: ‘something that is rare’. The idiosyncrasy of this reading can be seen in the absence of such a reading for *severity*, which cannot mean ‘something that is severe’—similarly for *obesity*, *sanity*, *clarity*, *purity*, *analyzability*.¹² Suppose we are correct in applying the sentence-level restriction on idioms to the word level. Then the reading in (57b) is assigned by a rule that applies only if *rare* and *-ity* are sisters.

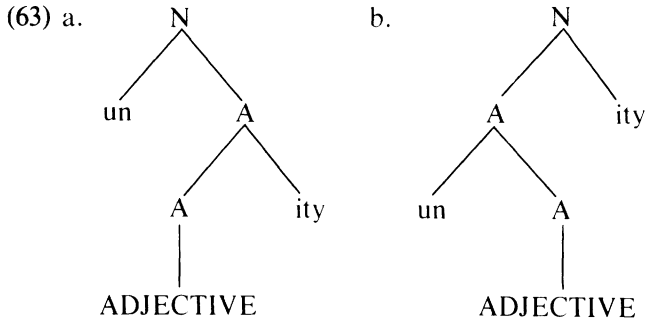
Now suppose we were to construct a word that contained the morphemes *rare* and *-ity*. Suppose these morphemes were sisters at S-structure, but at LF something forced them not to be sisters. If the rule of idiosyncratic interpretation applies to LF (or to a representation isomorphic to LF), then it will not be able to apply in this situation.

In section 2.3 we considered words of the type *un-ADJECTIVE-ity* and noted the conflict between Level Ordering at S-structure, which forces the adjective and *-ity* to

¹¹ Perhaps nonsisters X and C could participate in such a rule in (62b), where C is the head of B. This might be the case in idioms like *give NP the finger*, if we assume that *NP the finger* forms a constituent, as suggested by Kayne (1982; also see Stowell (1981; 1983)), of which the rightmost member is the head.

¹² On the other hand, like *rarity* are *impurity*, *surity*, *peculiarity* (cf. (58b)), and perhaps others. I am grateful to Armin Mester for pointing out the slight generality of the “*rarity* rule.”

be sisters, and c-selection at LF, which forces *un-* and the adjective to be sisters:



Level Ordering facts concerning stress, nasal assimilation, etc., argued for (63a). The evidence for (63b) was somewhat weaker, resting on the assumption that if *un-* c-selects (subcategorizes) a node, it is the unique sister of that node. So far, we have not seen any direct evidence for a structure like (63b). Nonetheless, if there are sisterhood requirements on both c-selection and idiosyncratic interpretation, and if there *is* a well-defined, independent level of LF, then we might find situations where these requirements conflict. Should we find such cases, we will have an argument for the sisterhood requirements and—most important—an independent, empirical argument for LF. Such cases do exist.

Consider a word like *unrarity*. This word is fairly acceptable in its normal, nonidiosyncratic sense, as in (64a), but completely unacceptable if we try to assign to *rare* plus *-ity* the idiosyncratic meaning we have been discussing, as in (64b):

- (64) a. ⟨We would have expected first editions of Hemingway to be hard to obtain.⟩
We were therefore shocked at the comparative *unrarity* of these priceless books.
- b. *A first edition of *SPE* is, however, a bibliographic *unrarity*.

Other examples are as follows:

- (65) a. ⟨Everyone had told us about how peculiar the Joneses were.⟩ But what shocked us about Bill Jones was actually his total *unpeculiarity*.
- b. ⟨Everyone had told us about how peculiar the IBM keyboard was.⟩ We were therefore shocked at its many **unpeculiarities*—for example, the location of its *Q* key.
- (66) a. The *reinstallation* of the elevator took a week.
- b. ⟨We were visiting the second nuclear reactor to be built on the site of the famous Northampton meltdown, which destroyed the first reactor.⟩ We were pleased to learn that smoking is prohibited on the grounds of the nuclear **reinstallation*.¹³

¹³ The prefix *re-* c-selects verbs, much as *un-* c-selects adjectives.

- (67) a. ⟨We have to depress his sodium level once and then redepress it.⟩ The *redepression* is the hardest part.
 b. ⟨Just when the country seemed to have climbed out of the Great Depression, there was a second depression.⟩ The **redepression* was hardest on those already unemployed.
- (68) a. Bill's *misrepresentation* of Mary's views went unnoticed in the press.
 b. ⟨Linguisticz (1982) thus gives an incorrect D-structure representation for the Moldavian sentence.⟩ Consider now another D-structure **misrepresentation*, seen in tree (215) below.¹⁴

What is crucial here is that the words under consideration are not formally ungrammatical, given the right context. They simply lack one particular reading—namely, the reading that depends on a rule of idiosyncratic interpretation.¹⁵ This is because there is nothing intrinsically wrong with an LF structure like (63b), with *rare* as the adjective. The rule of idiosyncratic interpretation, however, cannot apply since *rare* is not a sister of *-ity* in this structure.

Thus, if we take the (63a) structure to be well-motivated by phonological considerations (the Level Ordering Hypothesis), the facts just considered argue strongly for the independent existence of the (63b) structure, since with it we can explain the facts. In turn, since (63a) and (63b) are both subject to a *clustering* of rules, constraints, and principles, we have the classic argument for two levels of representation.¹⁶ The fact that some rule in a derivation (e.g. the so far undiscussed rule deriving (63b) from (63a)) yields

¹⁴ Some examples of this sort do occur—for example, the term *misgeneration*, which occasionally occurs in the linguistic literature. By and large, however, I find examples like (68b) bad. This is explained if *mis-* selects verbs, with occasional exceptions, such as *misfortune*. Then *misgeneration* can fall into this category of exception, *mis-* being the sister of the noun *generation* at all levels of representation.

¹⁵ An LI reviewer has correctly pointed out that we must establish the absence of a general principle forbidding affixation to lexical idioms. Some of the examples are a bit far-fetched, but they seem less ungrammatical than the uses of *unrarify*, *misrepresentation*, etc., discussed above. Thus:

- (i) ⟨You say Mary is a real rarity among semioticians?⟩ I think she's a *nonrarity*! [*non-* is not restricted to adjectives; cf. *nonmusician*]
- (ii) The location of the shift key on the IBM keyboard is only a *pseudo-peculiarity*.
- (iii) ⟨This is not a question of LF as a whole, but of individual LF representations.⟩ Such *representational* questions can only be decided by looking at other representations of the same sort.
- (iv) ⟨You say this is a nuclear installation?⟩ It sure doesn't look very *installation-like*.

¹⁶ A similar argument can be constructed that is relevant to the nature of allomorphy. Consider the allomorphy seen in pairs like *explain* ~ *explicable* vs. **explic* as a verb. Suppose the morpheme *explic* is a variant of *explain* with the same “bound” semantic properties as morphemes like *ceive*. Though it may mean *explain* in combination with *-able*, it has no meaning on its own, requiring a rule of idiosyncratic interpretation to interpret it. Hence, a word containing an occurrence of *explic* that is not a sister to *-able* will be ruled out by filter (54). Suppose the addition of a Level II prefix that selects verbs were to force *explic* and *-able* not to be sisters at LF—*re-* and *mis-* being examples. We predict that the rule of idiosyncratic interpretation that permits *explic* and similar allomorphs would be blocked. The facts bear out this prediction. Compare:

- (i) reexplainable vs. **reexplicable*
- (ii) ?redemonstrable vs. **redemonstrable*
- (iii) ?misperceivable vs. **misperceptible*
- (iv) reeducatable vs. ??reeducable

Thus, certain instances of allomorphy may find their solution at LF, providing another argument, similar to the previous one, for a distinction among levels.

an output distinct from its input does *not* argue for two interesting levels of representation. Rather, we need to see a *number* of significant properties attached to both output and input. But this is exactly what we have seen: *each* structure meets a number of independent conditions, and *each* structure has a collection of rather interesting properties; hence, we *are* dealing with two levels of representation. The proposed analysis differs in this way from other discussions of bracketing paradoxes, including those of Williams (1981a), Kiparsky (1983), and Moortgat (1984), who argue either for an essentially phonological rule of rebracketing (Kiparsky), for “corepresentations” at a single level (Williams; also see Williams (forthcoming)), or for powerful “type lifting” mechanisms at a single level (Moortgat).

Furthermore, by identifying two levels of representation we raise expectations of “modularity” that are not raised if we are dealing with some indifferent stage in a phonological derivation or with corepresentations. One of the two levels undergoes phonological rules, meets phonological conditions, and satisfies the language-particular morphological constraints of Level Ordering. The other satisfies categorial requirements, determines scope, is the level by which meaning—particularly idiosyncratic meaning—is directly determined. Were we to find some other associations of properties, we might well be surprised, but if we are dealing with two levels, a surface level and LF, we are not surprised.

4. Affix Movement and LF

The question to be answered now is this: Given that we are dealing with two levels of representation, what are the properties of the mapping from one level to the next?

4.1. *QR for Affixes*

In the previous section we adopted Lieber’s Percolation Conventions for morphologically complex words and saw how rules of idiosyncratic interpretation, aided by filter (54), allow the percolation properties of affixes to be derived from the fact that they bear subcategorization frames. We considered the suggestion that at both the word level and the sentence level only heads have subcategorization frames. If affixes are by definition morphemes with subcategorization frames, then they are heads. As noted by Selkirk (1982) and earlier by Williams (1981a), the percolation properties of affixes (Percolation Convention II) are exactly those of heads in syntax.

Suppose now that S-structures for words are formed much as sentential S-structures are in the theory of Chomsky (1981; 1982) and Stowell (1981). Any structure is allowed that meets the criteria for well-formedness, including the criteria in (46). We probably want to add to our criteria for both S-structure and LF something like (69):

(69) All branching is binary.¹⁷

¹⁷ There is an empirical argument for (69). Were it not true, we would expect to find affixes that demand, as a positional or categorial requirement of subcategorization, two sisters—a sort of affixal *deem*. For example, *A + B + affix* would be grammatical, where *A* and *B* are morphemes, but *A + affix* would be ungrammatical.

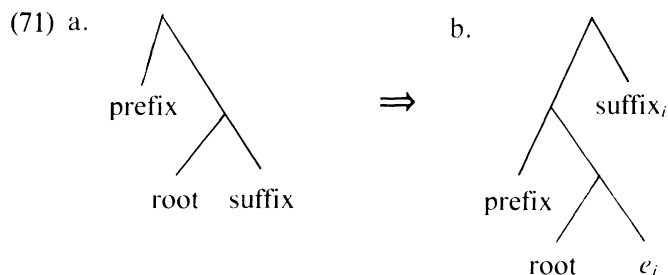
Now, how are word-structure representations at S-structure mapped onto LF representations? Suppose the mapping is accomplished by May's rule of QR, extended to affixes:

(70) *QR*

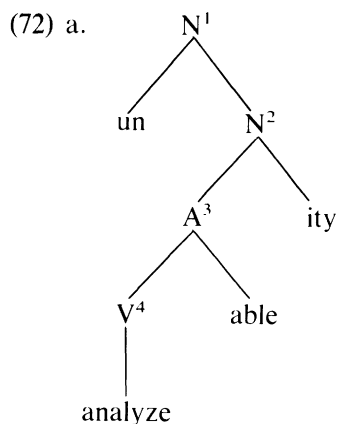
Adjoin a category C to some node that dominates C.

In this formulation, QR is no longer a rule that expressly applies only to quantifiers. Although *QR* was originally an abbreviation for *Quantifier Rule*, we will take the entire content of the rule to be what is given in (70). When we allow affixes like *-ity* to “undergo QR,” we will not be claiming that *-ity* is a quantifier (although comparative *-er* might be). We will simply be claiming that it is adjoining to a certain category by virtue of the movement rule described in (70).

Pushing the parallel between word-level and sentence-level mappings to the utmost, I suggest, speculatively, that movement of affixes under QR leaves a coindexed trace. I further suppose—though I shall soon revise this supposition—that LF representations are derived in one mapping from fully formed S-structure trees, where these trees are created as just discussed. Trees with the organization seen in (71a) are thus mapped onto trees like (71b) (where for the moment the nodes are not labeled):



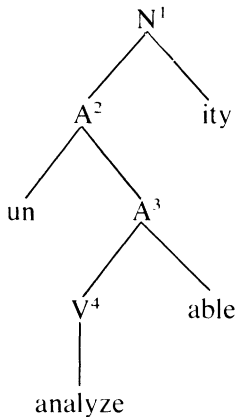
Now let us consider an actual S-structure—for example, the S-structure of *unanalyzability*. The Percolation Conventions that remain, plus the properties of affixal heads, give the following labeled tree:



The assumption that *un-* belongs to the “null category class” to which *counter-* belongs is now essential if we want to allow structures like (72a)—motivated by Level Ordering—at S-structure. In (72a) *un-* is sister to a noun (acceptable if c-selection does not apply until LF) and yields a noun, since it belongs to the categorially transparent null class. (72a) satisfies Level Ordering because *un-*, a Level II affix, c-commands all the Level I affixes, namely *-able* and *-ity*.

At LF, as we have seen, a structure like (72a) violates c-selection. Suppose the structure is saved by an application of QR, and suppose QR does *not* leave a trace. If the suffix *-ity* adjoins to the whole noun, it will leave *un-* as the sister of the adjective *analyzable*, and *-ity* as the sister of the adjective *unanalyzable*. C-selection by both these affixes will be satisfied:

(72) b. LF—no traces



It is crucial that *-ity* “gets out of the way” so that *-able*’s features can be percolated up to the node that is a sister to *un-*. Also, since only the movement of *-ity* allows c-selection to be satisfied at LF, this movement is effectively made obligatory. Nonetheless, we may suppose that QR, in a word as in a phrase, is formally *optional*. It applies obligatorily only when other principles force it to apply.¹⁸

4.2. If Affixes Leave Traces

Suppose now that QR of affixes *does* leave a trace. The same effect is required: for the purposes of percolation, trace must act as if it is not there. *Un-* must be a sister to a

¹⁸ In the Russian prefixed verbs, QR of inflectional affixes is forced in at least some cases by the need for prefix and verb to undergo a rule of idiosyncratic interpretation at LF. There is, however, a hole in our discussion: no independent principle forces movement of comparative *-er* in the cases considered in section 2.1. Recall that when *-er* interacts with *un-*, *-er* must have wide scope at LF. I have no explanation for this fact, except to note that it is similar to facts frequently observed with certain sentence-level quantifiers—for example, *each*. In this context, it is relevant to note, following an observation of I. Roberts (personal communication), that some speakers apparently accept combinations of comparative *-er* with *non-* (e.g. *nonhappier*), and in such cases it is *non-* that takes wide scope.

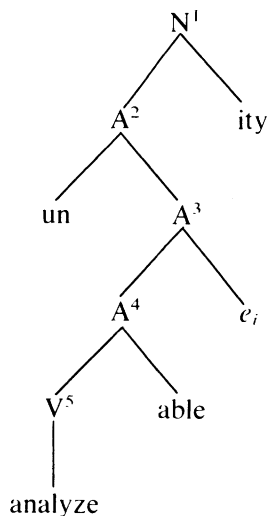
projection of the adjective *analyzable*. The following stipulation is therefore needed:¹⁹

(73) *Trace Stipulation*

The trace of an affix belongs to the null category class $\emptyset$

If we reapply QR to (72a), this time leaving a trace and obeying (73), which renders the trace transparent to percolation, we obtain the structure in (72c):

(72) c. LF—with traces



C-selection is satisfied by this new tree. After *-ity* moves, the sister of node 4 is a trace. By stipulation (73), traces have the null category specification. By Percolation Convention III, the features of node 4 percolate to node 3, assigning node 3 to the category A.

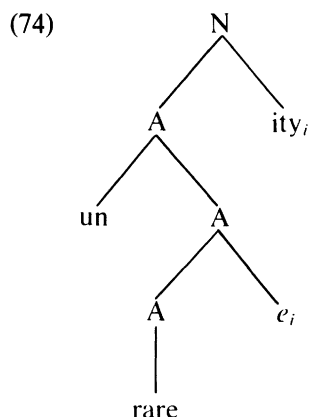
Since *un-* also belongs to the null category class, Convention III reapplies to percolate the features of node 3 to node 2. Hence, node 2 also receives features for the category A. The prefix *un-* is correctly the sister of an adjective, and the suffix *-ity* is also the sister of an adjective. C-selection is satisfied everywhere, as a result of applying QR to *-ity*.

Lieber's Percolation Conventions play an essential role here. It is also vital that category labels adhere to nodes *only* by virtue of the conventions. When the structure changes in a derivation such that the conventions would assign new labels to old nodes, the conventions reapply. The structure at each level is determined by the principles of

¹⁹ I give (73) the unfavorable title of "stipulation" because, if it is true, it surely must follow from some other part of the grammar. It might. N. Chomsky (personal communication) notes that the trace of an affix is plausibly neither an affix nor a root, and therefore is in fact expected to fall under Lieber's Percolation Convention III and not percolate its features. Also, since an affixal trace is not itself an affix, it might well not have any percolable features in the first place.

that level, and among the principles that independently characterize LF and S-structure are the principles behind the Percolation Conventions.

Another point, noted by M. Enç (personal communication), deserves mentioning. For the argument in section 2 concerning lexical idioms to hold, the trace of a moved affix must not be allowed to “stand in” for the affix itself in the idiom interpretation rule. We excluded the idiosyncratic reading of *rarity* from the word *unrarify* by noting that—if there are no traces—the LF structure for that word must be like (63b), where the adjective *rare* is a sister only to *un-*, and not to *-ity*. If there are traces, however, the LF structure for *unrarify* resembles (74):



In (74) *rare* is a sister to the trace of *-ity*, yet the rule of idiosyncratic interpretation fails. The trace thus fails to stand in for its antecedent in a rule of idiomatic interpretation.

This fact is surely troubling, but not shocking. There is a sentence-level rule whose effects are visible at S-structure and which resembles affixal QR: Heavy NP Shift. Heavy NP Shift is an S-structure rule at the phrasal level that has the properties of affixal QR and applies to nonquantifiers—adjoining them to a node that contains them.²⁰ It seems to me that this sentence-level “QR” also disrupts idioms:

- (75) a. ?John gave Bill *e* yesterday the answer.
 b. *John gave Bill *e* yesterday the finger. [* as an idiom]
- (76) a. ?John ate *e*, when Bill arrived, his worms.
 b. *John ate *e*, when Bill arrived, his words.

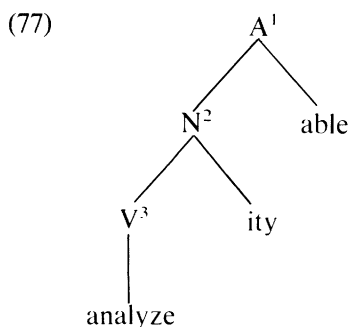
We have seen that the traces of affixal QR must be categorially transparent and cannot stand in for their antecedents under rules of idiosyncratic interpretation. One might therefore wonder, with good reason, if these traces exist. In section 5 I will suggest that there is a good argument for these traces. In particular, they help place interesting and apparently true restrictions on the mapping from S-structure to LF, and they are important in explaining a restriction on thematic roles in compounds.

²⁰ However, see Rochement (1978) for the suggestion that Heavy NP Shifted categories do have some sort of focus operator associated with them; cf. Stowell (1981).

First, however, let us consider in greater detail the action of our rule of QR. We will assume traces in what follows, but the discussion does not depend on them. In particular, let us consider some overgeneration by the system as it stands, and some ways to curb it.

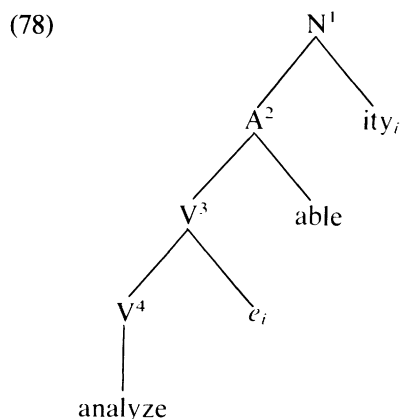
4.3. Overgeneration: Cyclic Mapping or String Vacuousness

We have assumed that QR applies freely to affixes. As things stand, nothing rules out “morpheme salads” in which c-selectional requirements are freely violated and LF movement makes everything turn out grammatical. For example, we have no way to exclude a word like **analyzitable* (*analyze* + *ity* + *able*), with an S-structure representation like (77):



We know that *-ity* c-selects an adjective and *-able* c-selects a verb. By hypothesis, this information does not matter at S-structure. (77) would only be ruled out at S-structure if Level Ordering or positional aspects of subcategorization were violated, which they are not.

Nor can we exclude **analyzitable* at LF, given our argument so far. If (77) is mapped unchanged onto LF, c-selection will of course be violated, since *-ity* is sister to a verb and *-able* is sister to a noun. Suppose we move *-ity*, adjoining it to node 1, just as we did with *unanalyzability*. The result would be (78):



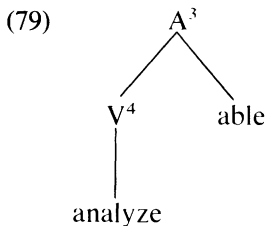
In (78) we have reapplied the Percolation Conventions, as required, after the operation of QR. C-selection is thus satisfied at the one level at which it is relevant, and there is no reason to mark **analyzitable* ungrammatical. We will discuss two solutions to this problem: a cyclic solution and a solution along the lines of Kiparsky (1983) and Pesetsky (1979).

4.3.1. A Cyclic Solution. One solution is suggested by the following observation. In the case of the grammatical *unanalyzability*, the categorial difficulties posed by the sisterhood of *un-* and the noun *analyzability* can be solved on the cycle at which *un-* is introduced. That is, at the cycle at which *un-* and *analyzability* are composed, the suffix *-ity* is already available to undergo QR and save the word from violating c-selection. By contrast, in **analyzitable*, no operation at the node 2 cycle (cf. (77)) can save this word. Not until *-able* has been introduced, on the higher node 1 cycle, is there any way to place *-ity* next to the adjective that it requires. Nothing but *-able* can save the constituent *analyze + ity*, and *-able* is introduced on a higher cycle.

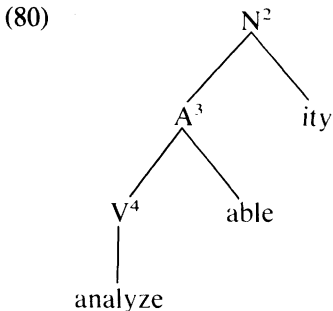
Let us exploit this difference between *unanalyzability* and **analyzitable* and see the theory of morphological S-structure-to-LF mapping that emerges. It will turn out that this theory is independently motivated.

We have argued that c-selection in morphology is relevant only at LF. If we exploit the difference between *unanalyzability* and **analyzitable* discussed above to explain the contrast, we must claim that c-selection at LF is checked cyclically, from the bottom up. If there is no way to mend a violation on the cycle at which it occurs, the word is ruled out.

Consider in this light the word *unanalyzability*, with the S-structure in (72a). C-selection starts with the innermost cycle:

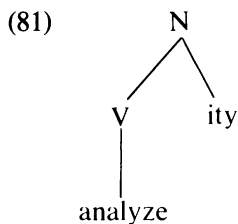


Since *-able* is sister to a verb, everything is fine. The node 2 cycle comes next:



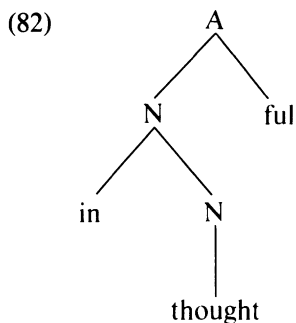
Since *-ity* is sister to an adjective, c-selection is still not violated. Problems arise for the first time on the next cycle—the structure in (72a). On this cycle, however, there is a way out: we can save the structure by moving *-ity*, as we have seen. C-selection is thus satisfied.

The nonword **analyzitable* can be ruled out on the first cycle:

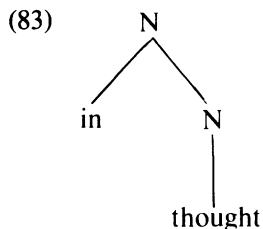


The suffix *-ity* is not next to the adjective that c-selection requires, and there is no way to save the structure on this cycle.

In similar fashion, we can exclude other morpheme salads like **inthoughtful*. Level Ordering here imposes the S-structure in (82), if *in-* is Level I and *-ful* is Level II:



Like *un-*, negative *in-* generally subcategorizes for an adjective (ignoring a few exceptions like *indecision*, discussed by Siegel (1971; 1974)). On the first cycle, c-selection at LF will examine the constituent in (83):



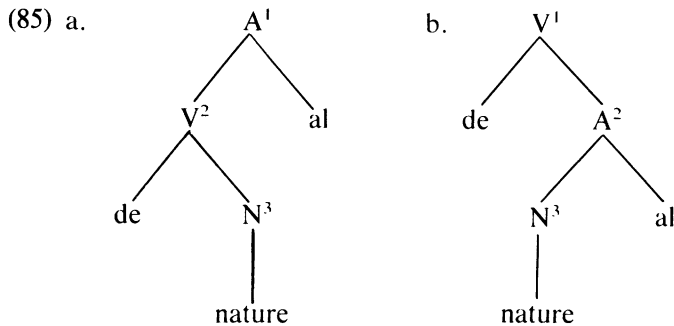
C-selection is violated in (83), and there is no way to satisfy it on this cycle.

Certain other cases of bracketing paradoxes will not arise, whenever the only way to save c-selection at LF for one affix entails violating it for another affix. Such a situation

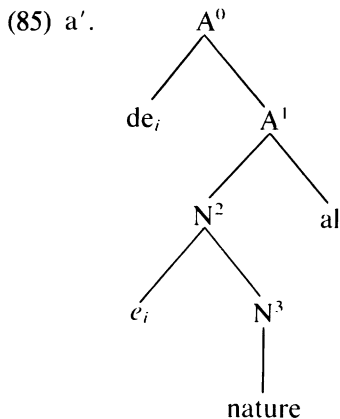
obtains in a number of cases discussed by Kiparsky (1983), e.g. **denatural*.²¹ The lexical entries for *de-* and *-al* are as follows:

- (84) a. *de*: V, [+ ____ N]
 b. *al*: A, [+ N ____]

From (84) it follows that both the adjective *natural* and the verb *denature* are possible words. Without even asking about the Level Ordering properties of *de-* and *-al*, we can see that any S-structure bracketing for **denatural* will lead to a violation at LF:

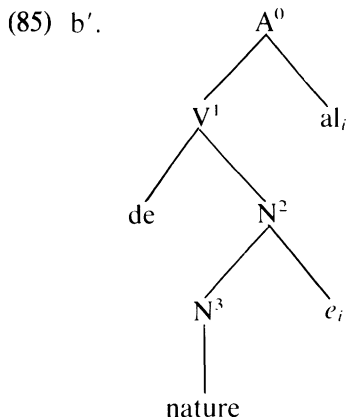


Consider first (85a). Clearly, c-selection is satisfied on the node 2 cycle but fails on the node 1 cycle, since *-al* selects a noun. The only option is to move *de-* by QR, yielding (85a'). After this movement, c-selection is satisfied for *-al* but is now violated for *de-*:



In (85b) c-selection is satisfied on the node 2 cycle, but it fails on the node 1 cycle, since *de-* selects a noun. The only solution is to make *-al* undergo adjunction to node 1, yielding (85b'):

²¹ The "cyclic" proposal here also recalls Kiparsky's idea that "subcategorization" (c-selection) holds both before and after his morphological rebracketing.



Now *de-* is a sister to the required noun, because of the Percolation Conventions, but the c-selectional properties of *-al* are not satisfied.

Now we face two questions: (a) Why is c-selection checked and QR applied cyclically? (b) Is there independent evidence for such cyclicity in the mapping to LF?

To answer the first question first: In Pesetsky (1979) I argued that cyclic application of certain phonological rules follows from the basic structure of the lexicon. In particular, I suggested that rules of cyclic phonology apply in the lexicon after each application of an affixation rule. As words are built up from the innermost root by successive concatenation of affixes, so the rules of phonology that apply in the lexicon will reapply in succession to each new domain created by the affixation—hence the cyclic effect.

Cyclic application of rules and principles in LF can be derived in the same way. If LF is built up in tandem with affixation in the lexicon, and LF principles checking c-selection apply—like the phonological rules of the lexicon—each time a new morpheme is added to the representation, then the cyclicity of c-selection follows immediately. Thus, in a word like *unanalyzability*, the affixation of *-ity* to *analyzable* yields an interim LF representation like (80), which must satisfy c-selection, and does. The further affixation of *un-* yields an LF representation like (72a), which can be saved by QR on that cycle, as we have seen.²²

Further, if we develop a theory of LF representations for words that is consistent with the model of Pesetsky (1979) or with further developments in Lexical Phonology, then the cyclic building up of LF is not only plausible, but necessary. These models presume that as words are built up at the level that we are now calling S-structure, internal bracketing is systematically eliminated by a variant of SPE's *Bracket Erasure Convention* (Chomsky and Halle (1968)). The actual S-structure of a fully formed word like *unanalyzability* will contain none but the most external brackets, as the result of

²² This "rule-to-rule" hypothesis recalls parallel proposals in Montague Grammar for syntax; see, for example, Partee (1975).

successive erasure of internal brackets. Nonetheless, as noted in Pesetsky (1979), semantic compositionality requires the existence of a parallel representation in which *un-analyzability* does have a rich internal structure (whose terminal symbols are perhaps not phonological matrices, but semantic representations of some sort). Since this rich structure cannot be recovered from an S-structure representation in which all bracketing and differentiation among morphemes has been eliminated, it must be built up in the course of affixation. In particular, no erasure of bracketing (or at least no wholesale erasure) must take place in LF.

For the proposed analysis, this necessity is even more obvious. LF cannot contain a Bracket Erasure Convention of the type envisaged for S-structure if syntactic operations like QR are allowed to affect affixes contained in previously passed cycles. If the boundary between *analyzable* and *-ity* does not exist on the cycle containing *un-*, then QR cannot analyze the suffix *-ity*, and there is no way for c-selection to be satisfied or even checked. Thus, if bracket erasure takes place at S-structure, then LF and S-structure *must* be built up in tandem, with LF snatching structure from the jaws of the Bracket Erasure Convention.

We have now answered the first question. A model is readily available in which the cyclic application of c-selection follows naturally, and this model is in fact forced upon us by the Bracket Erasure Convention of Lexical Phonology.²³

As for the second question, Muysken (1981) presents independent evidence for a cyclic derivation of word-internal LF representations. The evidence comes from Quechua dialects of central Peru. Muysken assumes that associated with argument-taking morphemes in their LF representations is something like what Stowell (1981) later called a *thematic grid*. The affixation of certain morphemes can modify this grid, coindexing arguments, adding and possibly subtracting arguments and predicates. In particular, associated with the reflexive morpheme *-ku* is a rule like (86):

(86) *Quechua Reflexive Rule*

$V - ku \Rightarrow x_i (V \dots y_i$

That is, "Bind the subject of the predicate expressed by the verb in the domain of *-ku* with an anaphor."

If a verb like *riku-* 'see' has a grid of the form $[x (V y)]$, where x is the "external argument" in the sense of Williams (1981b), then attaching the reflexive suffix *-ku*, as in (87),

²³ This analysis, however, undermines some strong motivation for the Bracket Erasure Convention: the Adjacency Condition for c-selection of Siegel (1977) and Allen (1978). In a model in which c-selection does not apply at LF, erasure of internal brackets in the morphology at the end of each cycle ensures that subcategorization frames can refer only to the outermost layer of bracketing. If c-selection is enforced at a level at which brackets are not erased, the Adjacency Condition for c-selection must be derived from some other principle—for example, government. We might still motivate Bracket Erasure for other instances in which the Adjacency Condition is relevant—for example, for features like [+Learned] in French phonology (Dell and Selkirk (1978)). Nonetheless, we must obviously use caution if we employ Bracket Erasure to strengthen an argument whose result is to undermine the motivation for Bracket Erasure.

- (87) riku – ku – n
 see refl 3

yields a grid of the form $[x_i (V y_i)]$. *Riku-ku-n* thus means ‘he sees himself’.²⁴

Associated with the causative morpheme *-chi* is a rule like (88) (slightly modified):

- (88) *Quechua Causative Rule*

$V - chi \Rightarrow x \text{ (CAUSE (... V ...))}$

That is, “Embed the predicate expressed by the verb in the domain of *-chi* as an argument to a causative predicate.”

The verb *riku-chi-* will thus have a grid of the form $[z \text{ (CAUSE (} x (V y) \text{))}]$ and will mean ‘cause to see’.

The argument for cyclic construction of LF comes from verbs formed using both the reflexive morpheme and the causative morpheme. In the Quechua dialects that Muysken considers, these morphemes may be affixed to a verb in either order. Thus, from the verb *mayla-* ‘wash’, we may derive the two verb forms in (89). What is interesting is how the order of affixation affects the interpretation:

- (89) a. mayla – chi – ku – n
 wash CAU refl 3
 ‘he_i causes someone to wash him_i’
 b. mayla – ku – chi – n
 wash refl CAU 3
 ‘he_i causes someone_j to wash himself_j’

Only in (89a), where the causative *-chi* has been affixed before reflexive *-ku*, does reflexive binding involve the subject of CAUSE. Muysken notes that this follows straightforwardly if LF is built up cyclically, and if the reflexive and causative rules must apply on the cycle in which the relevant morphemes are introduced. In (89b), at the level at which the reflexive rule must apply, there is no higher causative predicate; hence, the reflexive binding rule cannot involve this predicate. Schematically:

- (90) a. maqa chi ku n
 cycle 1 _____ $x \text{ (HIT, } y)$
 cycle 2 _____ $z \text{ (CAUSE (} x \text{ (HIT, } y) \text{))}$
 cycle 3 _____ $z_i \text{ (CAUSE (} x \text{ (HIT, } y_i) \text{))}$
 cycle 4 _____ $3_i \text{ (CAUSE (} x \text{ (HIT, } 3_i) \text{))}$
 b. maqa ku chi n
 cycle 1 _____ $x \text{ (HIT, } y)$
 cycle 2 _____ $x_i \text{ (HIT, } y_i)$
 cycle 3 _____ $z \text{ (CAUSE (} x_i \text{ (HIT, } y_i) \text{))}$
 cycle 4 _____ $3 \text{ (CAUSE (} x_i \text{ (HIT, } 3_i) \text{))}$

²⁴ Muysken notes (p. 454) that *-ku* may also have a medial or mediopassive interpretation, which we will ignore here. He also notes (personal communication) certain complications with other affixes.

Muysken's analysis (also see Baker (1983)) thus strongly supports the cyclic construction of word-internal LF representations for words and the cyclic application of principles of LF.

To sum up this section: we have seen that the cyclic theory espoused here is necessary in any case to handle the overgeneration of nonwords by QR. This theory is also required by the Bracket Erasure Convention in S-structure morphology and is supported by evidence from Quechua reflexives and causatives.

4.3.2. The Cyclic Solution vs. Rebracketing. What is meant by the term "movement rule"? Movement rules in generative grammar may be diagnosed by one of the following criteria:

- (91) a. a change in dominance (hierarchical) relations among terminal or non-terminal elements;
- b. a change in a string of terminal or nonterminal elements.

The earliest transformational studies emphasized (91b). Changes in hierarchical structure were incidental and predictable by-products of changes in a string. In more recent work, these issues have become less clear. For example, Chomsky (1981) suggests an important role for chains of positions in a phrase marker in defining Move α , and he proposes for some languages a rule of the form *Assume Grammatical Function*, which is not stated in terms of strings at all.

The discussion so far has argued for a level of LF by reference to differences in hierarchical relations between well-motivated S-structures and LF representations. These are, of course, expressible as differences in nonterminal strings (strings not entirely composed of terminals). For example, the string (*un*, N) is part of the derivation of the S-structure for *ungrammaticality*, but not of the LF. Are there ever distinctions between the *terminal* strings at S-structure and at LF? Surprisingly, this is an empirical issue that is relevant to a difference between the theory presented here and certain other theories of bracketing paradoxes in word structure.

Let us consider for simplicity's sake the variant of affixal QR that does not leave traces. As noted earlier, Pesetsky (1979) and Kiparsky (1982) proposed that bracketing paradoxes of the sort we have considered be handled by means of a "rebracketing rule." A rebracketing rule is a rule that, like our QR, alters the dominance relations in a tree. The node labeling after rebracketing is presumably adjusted much as in the proposal made here, though neither Pesetsky (1979) nor Kiparsky (1982) is explicit about this. What distinguishes a rebracketing rule from our QR is a further restriction:

- (92) *String Vacuousness Restriction*

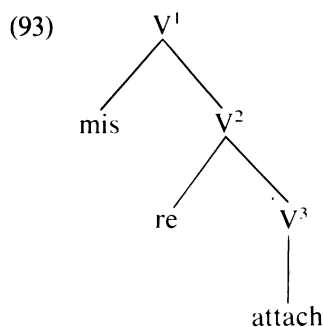
The terminal string output does not differ from the terminal string input.

Once rebracketing is seen, appropriately, as QR plus the string vacuousness re-

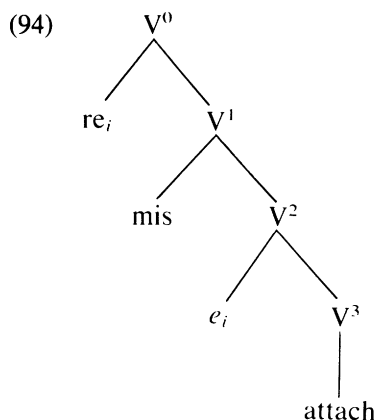
striction in (92), it seems conceptually nonoptimal.²⁵ Nonetheless, some empirical evidence may support such a restriction.

In each of the examples so far, QR has involved the movement of right-peripheral material to a position where it c-commands a prefix. An examination of the derivations presented will show that in no case does the output terminal string necessarily deviate from the input terminal string. Can QR apply when the output terminal string *must* differ from the input—when a suffix must move *over* another suffix, or a prefix must move over another prefix?

Some of the data are mixed and abstruse. By and large, however, it seems that QR cannot apply in the cases I have been able to develop, or that it applies only marginally. For example, consider a verb with two non-category-changing prefixes:



In its most natural reading, this verb is interpreted compositionally to mean ‘incorrectly [attach for the second time]’. That is, if something was correctly attached the first time, but should not be attached twice, then someone who did in fact attach it a second time would be said to have *misreattached* it. If there is no string vacuousness constraint (rebracketing), nothing prevents optional adjunction of *re-* to V¹ on the last cycle, yielding (94):



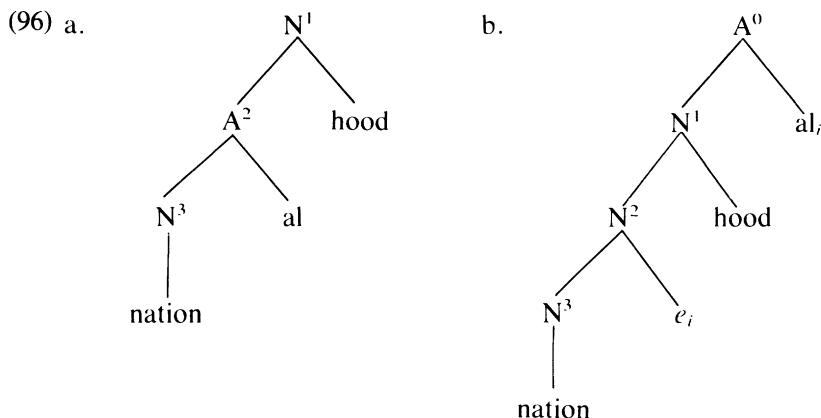
²⁵ Perhaps the rebracketing proposal can be shown to be natural from some broader perspective. For example, Sproat (1984) suggests that a version of rebracketing might derive from principles of linking in autosegmental phonology.

Structure (94) would be interpreted to mean 'for a second time [attach incorrectly]'. To be sure, this meaning is not inconsistent with the meaning that follows from (93) (as R. Sproat has noted (personal communication)), but it is distinct. If (94) is an acceptable structure, there should be nothing wrong with a discourse like (95):

- (95) ⟨Bill misattached the antenna once.⟩ Then, after checking the instructions, he went back and mis-*re*-attached it!

My own intuitions are that (95) is marginally allowable, with heavy contrastive stress on *re*-, but many other speakers disagree.

A clearer case, which supports the string vacuousness restriction, is noted by an LI reviewer. Consider an affix like *-hood*, which attaches to nouns and yields nouns, and an affix like *-al*, which attaches to nouns and yields adjectives. Now consider a putative word like **nationalhood*, with the S-structure in (96a) and the LF representation in (96b):



C-selection is satisfied on the N^2 cycle. On the N^1 cycle, a problem arises, since *-hood* is sister to an adjective before QR. This problem is solved on that cycle, however, by adjoining *-al*, as in (96b). Since the word is in fact ungrammatical, and the putative movement does violate a string vacuousness restriction, this evidence may support that restriction.²⁶

Consider also cases in which bracketing paradoxes arise at the NP level. It may be that the scope of affix movement is limited not to the word, but to the phrase. Thus, consider well-known cases of syncategorematic adjectives like *transformational* in *transformational grammarian*. In this example, the semantics supports a structure in which *transformational* and *grammar* are sisters and undergo a rule of idiosyncratic interpretation. This sisterhood (or near sisterhood, if there are traces) could only be achieved if *-ian* underwent QR and moved out of the way.²⁷

²⁶ The category change at the word level in these examples is not crucial, since the addition of a final *-ness* to keep the category constant does not alter the ungrammaticality of the example: **nationalhoodness*.

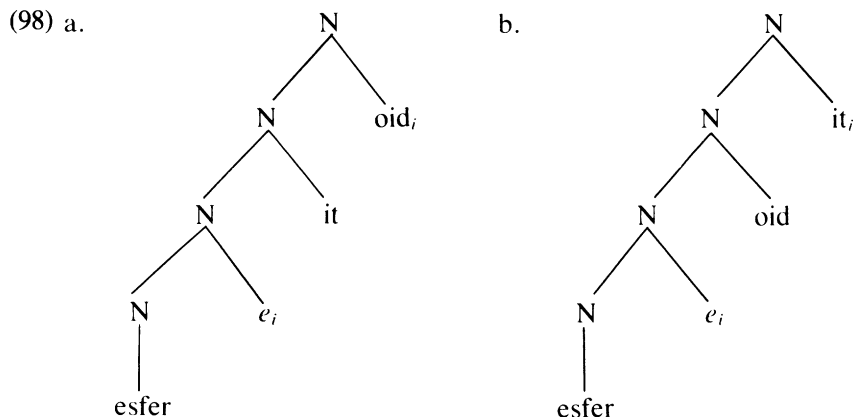
²⁷ Also compare phrases like *unhappier with us*, noted by Fabb (1984). He suggests that the movement of *-er* motivated in section 2 of this article be extended to the phrase level. See also the discussion in section 4 of phrasal θ -marking by the heads of compounds.

Now consider **transformational tall grammarian*.²⁸ The judgment of ungrammaticality is, I think, quite solid. In order for *transformational* and *grammarian* to be sisters here, the adjective *tall* would have to undergo QR and move out of the way. We clearly want this movement to be excluded. The string vacuousness prohibition would do so correctly.²⁹

A somewhat different conclusion is suggested by Spanish diminutives. O. Jaeggli (personal communication) reports that the following Latin American Spanish forms are both ambiguous:³⁰

- (97) a. ?esfer – oid – ita
 sphere sort-of dimin.
 b. ?esfer – it – oide
 sphere dimin. sort-of

The most natural interpretation of each form does correspond to its S-structure constituency. (97a) most naturally means ‘a cute little spheroid’, and (97b) means ‘almost a cute little sphere’. Nonetheless, it seems that each word may have an interpretation in which the scopes of *-oid* and the diminutive are reversed, which is expected if QR applies to the inner affix:



²⁸ I am grateful to D. Steriade for pointing out these examples to me.

²⁹ Before leaving this topic, we should consider another possibility. Walinska de Hackbeil (1983, forthcoming) suggests that the X-bar theory notion of *specifier* might play a role in morphology. Her examples distinguish two uses of the suffix *-ish*, a specifier-like usage in *reddish* and a nonspecifier usage in *boyish*. Suppose that prefixes like *re-* and *mis-*, as well as adjectives in NP, are considered “specifiers” in this sense (perhaps “noncomplement” and “nonhead” are more accurate here). In syntax, such “specifiers” also seem not to show scope ambiguities: thus, *not more happy* and *more not happy* are not potentially synonymous. Perhaps our “string vacuousness restriction” then really involves some constraint on moving specifiers. If prefixes and Walinska de Hackbeil’s *-ish* are specifiers, B. Levergood (personal communication) notes that words (?) like *irrelevantish*, *unrelevantish*, *impracticalish*, *unpracticalish* should be ambiguous, given the different S-structure bracketings predicted by the differing Level membership of *un-* (II), *in-* (I), and *-ish* (II).

³⁰ C. Otero notes that a gender distinction may be responsible for the failure of both these examples in European Spanish.

Thus, there is some evidence in favor of adopting a string vacuousness restriction on word-level QR, but Spanish raises a problem with this. I hesitate to constrain QR in this way, since the prohibition does not naturally derive from other properties of the grammar, nor is it motivated elsewhere. Indeed, the contrary restriction, prohibiting string vacuous movement, has occasionally been proposed in the syntactic literature. Also, if the string vacuousness proposal is correct, there will be two explanations for the English examples that motivated cyclic construction of LF, since saving a word like **analyzitable* from c-selection as in (78) involves a violation not only of cyclic checking of c-selection but also of the string vacuousness prohibition. This also argues against the string vacuousness proposal, in view of the Quechua evidence for cyclicity. I therefore conclude by tentatively adding the string vacuousness prohibition to the proposed system, but noting the conceptual and empirical problems with this move.

5. Traces of Affixes, Thematic Roles, and the Binding Theory

The analysis I have proposed has four basic parts, of which the fourth has been spelled out but not yet supported:

- A. Words have representations at more than one level, and these representations are not necessarily isomorphic.
- B. One of these levels is LF, with the properties discussed in section 1.
- C. The mapping from morphological S-structures to LF representations involves syntactic movement, in particular a rule like QR (perhaps with a string vacuousness restriction).
- D. This movement leaves traces.

Though the lower claims in this list are contingent on the claims higher up, the reverse is not true. Thus, it is possible that claim A is true, while claims B, C, and D are not, or that claims A and B are true, while claims C and D are not. So far we have argued for the first three claims explicitly, leaving claim D a “matter of execution.” Claim D is the trickiest, since, as we have seen, traces of affix movement, if they exist, must be categorially transparent (i.e. belong to the null category class) and must not be required to satisfy c-selection. Nonetheless, affixal traces, if they exist, are of great interest, for familiar reasons (concerning knowledge about empty categories; see Jaeggli (1982), Chomsky (1981), Bouchard (1984)).

In this section I argue for affixal traces by showing that a generalization about English compounding discussed by Selkirk (1982) is explained if they do indeed exist.

5.1. *Traces and the Binding Theory*

I cautioned earlier that our “QR” does not single out quantified NPs and that its application is not dependent on quantification in any way. Indeed, in (75)–(76) I noted that the English phenomenon of Heavy NP Shift may be an instance of “QR” applying to a nonquantified NP: it also involves adjunction of an element to a category that contains

it. I argued that the failure of affixal traces to “stand in” for their antecedents in word-internal rules of idiosyncratic interpretation (idioms) has a parallel in the failure of idioms to be preserved under Heavy NP Shift.

Arguments of this sort are potentially good arguments for the existence of affixal traces, since if we can explain a phenomenon by reference to general properties of certain objects, then these objects—which play a crucial role in the explanation—might very well be real. In the former case, of course, a simpler solution might eliminate the traces completely (as noted), since their failure to “stand in” could be explained by their failure to exist.

A similar situation arises in sentence syntax. The A-bound empty categories *NP-trace* and *PRO* fail to block *wanna* contraction, are invisible to the Case Filter, and generally fail to give much empirical proof of their existence. Nonetheless, there is at least one powerful argument for the existence of these A-bound empty categories: that the locality restrictions on A-binding can be explained by the same binding restrictions (in particular, principle (A) of the binding theory of Chomsky (1981)) that are independently motivated for certain overt anaphors. Because of the power of this explanation, it has been fruitful to assume the existence of these A-bound categories and seek explanations for the cases in which they do not act as expected.

Similarly with affixal traces, I believe. We are willing to look for independent explanations of their failure to manifest themselves for c-selection and percolation, if other phenomena can be explained by independently motivated constraints on their distribution. In particular, it seems that both affixal traces and Heavy NP Shift traces group with NP trace in displaying the properties of anaphors with respect to the binding theory.³¹ Let us consider Heavy NP Shift first. Compare:

- (99) a. John [_{VP} [_{VP} gave e_j to Mary] *copies of his latest book_j*]
 b. *John_i [_{VP} [_{VP} was believed [_S e_i to have bought e_j] by Mary] *copies of his latest book_j*]

In both examples, we are interested in the traces indexed “j.” (99b) is constructed to ensure that Heavy NP Shift has moved the NP *copies of his latest book* out of the domain of the subject of its sentence. By placing the NP to the right of a *by*-phrase that belongs to the higher sentence, we ensure that the NP belongs to that higher sentence. We can explain the ungrammaticality of (99b) if we assume that the trace of Heavy NP Shift is subject to principle (A) of the binding theory. This principle incorporates a version of the older Specified Subject Condition; an anaphor in the position of e_j in (99b) would be free in the domain of the nearest subject and would be ruled out.³²

³¹ Another phenomenon that involves QR-like adjunction may be clitic movement in Romance languages; see Kayne (1975) and subsequent literature for arguments that the traces left by clitic movement also have the binding properties of anaphors.

³² If the trace of Heavy NP Shift acts like an anaphor, then the definition of anaphor in Chomsky (1981), which depends on the distinction between A- and $\bar{A}$ -binding, needs revision, since Heavy NP Shift involves A-binding. One possibility, deriving from Borer (1982), is that the distinction relevant to the definitions of variable and anaphor is not A- versus $\bar{A}$ -binding, but Operator versus non-Operator binding. The trace of Heavy NP Shift may be $\bar{A}$ -bound, but its binder is not an operator. Therefore, if we revise Chomsky’s system appropriately, it will act like an anaphor.

Returning to word structure, let us ask how we can discover whether the trace of a moved affix, like the trace of Heavy NP Shift, has the properties of an anaphor. One sign might be that it cannot be free in the domain of a subject. The argument proceeds in three steps.

5.2. Step One: θ -Roles in Deverbal Compounds

Consider compounds like *tree eating*, *tree eater*, *moth-eaten*.³³ The category of the compound is determined by the second word of the compound, so that *moth-eaten* is an adjective because *eaten* is an adjective, whereas *moth eater* is a noun because *eater* is a noun. This may simply arise from something like Williams's (1981a) Right-hand Head Rule for English (as an LI reviewer reminds me). For our purposes, we may simply note that the first member of a deverbal compound is transparent for percolation purposes and acts like a member of the null category class.

With these preliminaries out of the way, I will argue that under certain circumstances the first element in an English deverbal compound may receive a θ -role from a lexical head in much the same way a sentence-level argument does.

Example (100) shows that there is θ -marking in compounds:

- (100) a. book reading
 b. reading of novels
 c. *book reading of novels

If we assume that the θ -role normally assigned by *read* to its subcategorized complement is assigned to *book* in (100a) and to *novels* in (100b), then we explain (100c) as a violation of the θ -Criterion, which forbids assigning the same θ -role to two arguments.³⁴

Is θ -marking obligatory in compounds, as it is at the sentence level? Consider the deverbal compound *tree eating*. Selkirk (1982) (cf. also the *First Sister Principle* of Roeper and Siegel (1978)) notes that this compound is ambiguous: on one reading, *tree* bears a thematic relation that is normally assigned to the direct object of the transitive verb *eat* at the sentence level; on another reading, *tree* functions as a locative of the intransitive verb *eat*. Thus, the sentence *tree eating is discouraged* may be paraphrased as either (101a) or (101b):

- (101) a. Eating of trees is discouraged.
 b. Eating in trees is discouraged.

³³ We probably wish to exclude the ternary branching structures for these compounds suggested by Kiparsky (1983), for the reasons discussed in connection with principle (69). Empirical evidence from coordination (also noted by Strauss (1983)) supports a major constituent boundary between the first and second orthographic words of compounds, but nowhere else: *tree and bush eating*, *tree eating and munching*, **tree eat- and munching*, **tree eat-ing and -er*. Fabb (1984) notes that a ternary structure can account for these facts, but only if an additional phonological stipulation is made for the last two cases, and if the second is derived from *tree eating* and *tree munching* by deletion under identity. The scope of word-internal deletion under identity, if it exists, is subject to restrictions, however, which I will not go into here.

³⁴ A more specific account of this paradigm follows from the theory of Lebeaux (1984), which I touch on in section 4.5.

Let us now consider these paraphrases themselves in greater detail. They show deverbal forms in *-ing* with phrasal, rather than compound-internal, complements. Selkirk makes an interesting observation about such cases, which I rephrase slightly in (102), illustrated in (103):

- (102) If a deverbal form is part of a compound, and if phrasal complements are added to the compound, directly θ -marked (subcategorized) arguments of the head of the compound must be found *inside* the compound.
- (103) a. Pasta eating in trees is discouraged.
 b. *Tree eating of pasta is discouraged.

We will assume for now that the internal θ -role “theme” assigned to *pasta* in (103a–b) is directly assigned by the verb *eat* (ignoring the role of the dummy Case assigner *of*). On the other hand, locative roles of the type assigned to *trees* in (103a) are not assigned by the verb, but either are assigned by the preposition *in* or are intrinsic, as in the locative reading of *tree eating*.

The verb *eat* has both a transitive and an intransitive use. In the transitive use, it directly assigns an *internal* θ -role (Williams’s (1981b) terminology); in the intransitive use, it does not. (103a–b) show the transitive use. In (103a) the nonsubcategorized locative is found outside the compound, and the directly θ -marked, subcategorized internal argument of *eat* is correctly found inside the compound. In (103b), however, the non-subcategorized locative *tree* is inside the compound, while the internal argument *of pasta* is outside the compound, violating Selkirk’s observation. Finally, in the simple phrase *tree eating*, where *tree* is interpreted as a locative (the (101b) reading), Selkirk’s observation also holds, since the *intransitive* version of *eat* does not directly assign any internal θ -role at all. Where *tree* is interpreted as theme, it is inside the compound, trivially satisfying Selkirk’s observation.

Selkirk’s observation, in our interpretation, can be explained rather simply by means of Chomsky’s (1981) θ -Criterion, but we must be very specific about how this θ -Criterion works. The θ -Criterion makes obligatory a biunique assignment of θ -roles to arguments. This assignment, however, must be contingent on the independent existence of the position that is to receive a given θ -role. Specifically:

If a structural position that can be θ -marked is obligatory, then it is obligatorily θ -marked by an element that may θ -mark it; if such a position is only optionally present, then θ -marking of this position is correspondingly optional, and will apply just so as to satisfy the θ -criterion. (Chomsky (1981, 40))

More generally, the presence of a position that can be θ -marked implies that this position *must* receive an available θ -role. If the position is independently obligatory, then it must be present and must be θ -marked. Examples of independently obligatory positions at the sentence level are subcategorized positions and the subject of S.

Obviously, nothing *forces* a deverbal noun to participate in a compound. If a deverbal noun has a θ -role that could be assigned to the first member of a compound, it does not have to be assigned, since there does not have to be a compound in the first place. If, however, a deverbal noun in a particular structure does happen to participate in a compound, then a position is *in fact* present that can receive a θ -role. If there is a θ -role to be assigned, it appears that this position must receive that θ -role, as we shall now see.

Consider first the transitive version of *eating*. Nothing forces *eating* to participate in a compound and to assign its θ -role of theme to the first member of such a compound. Hence, simple phrases like *eating of trees* in (101a) are grammatical. However, following Chomsky's discussion of the θ -Criterion, if *eating* does participate in a compound, and if the first member of a compound is a structural θ -position, then once a compound is built up, this first member must be θ -marked by *eat*.

In (103a), *pasta eating in trees*, there is a compound, and *pasta* is θ -marked with the theme role by transitive *eat*. *Trees* receives its θ -role from *in*, and the θ -Criterion is not violated. In (103b), **tree eating of pasta*, there is again a compound, and the presence of the phrase *of pasta* makes it clear that we are dealing with the transitive version of the verb *eat*. Given that the transitive verb *eat* has a θ -role to directly assign, and that the first member of a compound is a θ -position, then the first member of the compound must receive the θ -role that can be assigned to it. This means (a) that it cannot receive the locative role, and (b) that it is competing with the NP following *of* for the direct θ -role. By the θ -Criterion, both the first member of the compound, which contains an argument *tree*, and the phrasal object *pasta* must be assigned the direct θ -role, since both are arguments in a θ -position. By the θ -Criterion, the same θ -role cannot be assigned twice, so the structure is ruled out.

Finally, intransitive *eat* has no direct θ -role to assign, so *tree eating* receives the locative interpretation. *Eating of trees* is ruled out with this version of *eat*, which is no problem, since it is grammatical with the transitive version. Also, *eating in trees* is of course acceptable.

Thus, the first position of a deverbal compound is potentially a θ -position, and the θ -Criterion holds of it. Selkirk's observation is thus explained. But is the first member of a compound really an ordinary θ -position, requiring no further comment?

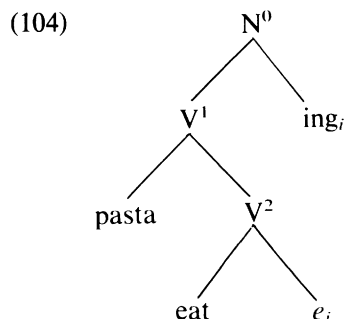
5.3. Step Two: θ -Roles and Adjunction

At the phrasal level, when an element receives a θ -role from V, it is normally a sister to a projection of that V.³⁵ Compare this to the situation inside the compound examples so far. In the S-structure for *pasta eating*, the argument *pasta* is not a sister to any projection of *eat*. Rather, it is a sister to a projection of *-ing*. Suppose a sisterhood

³⁵ See Lebeaux (1984) on the relation between θ -theory and X-bar theory.

condition on θ -role assignment holds inside compounds. How might we make *pasta* into a sister of a projection of *eat*?

By now, the answer will be unsurprising. Adjunction of *-ing* to the compound as a whole will place *pasta* next to a projection of *eat* at LF:



As required by the Percolation Conventions, including those governing affixal traces, nodes 1 and 2 are of category V. *Pasta* is now a sister to a projection of V and can receive a θ -role. Finally, *-ing* continues to be a sister of V—both before and after adjunction. Its c-selectional requirements are satisfied.³⁶

5.4. Step Three: Evidence for Traces

I now argue for traces. The evidence is an explanation for another generalization suggested by Selkirk. To prevent confusion with “Selkirk’s observation” rephrased in (102), I will label this *Selkirk’s Generalization* (Selkirk (1982)). This generalization is also a descendant of Roeper and Siegel’s First Sister Principle:

(105) *Selkirk’s Generalization*

The first member of a deverbal compound may not bear a relation to the second member which is normally expressed by the subject.

In the terminology of Lexical-Functional Grammar, which Selkirk adopts, the first member of a compound may not bear the SUBJ relation.

The data supporting this generalization fall into two categories:³⁷

- (106) a. *Weather changing is typical of western Massachusetts.
 b. *Time elapsing worries philosophers.
 c. *Heart beating is a good sign of life.

³⁶ We must suppose, however, that there is nothing wrong with *pasta* being a sister to *eating* at S-structure in apparent violation of the Projection Principle. Perhaps the Projection Principle merely requires the argument *tree* to be in a subcategorized position (sister to some lexical head) at S-structure and only cares whether that head in fact assigns a θ -role at LF, in the spirit of our suggestions concerning c-selection.

³⁷ Compound stress—main stress on the first element of the compound—is vital to these judgments, since many of the first members in these compounds can be used adjectivally. I am grateful to a number of people, including an LI reviewer, for forcing me to face the idiolect variation described below.

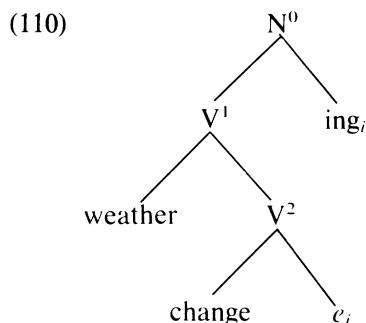
- (107) a. %Dog barking can be eliminated with a muzzle.
 b. %Child dancing is forbidden.
 c. %Student swimming is allowed only on Thursdays.

As indicated, all informants seem to agree on (106a–c), whereas there is variation in (107a–c). The reason may be connected to the following observation: all of the examples in (107), but none of the examples in (106), allow fairly acceptable paraphrases in which the first member of the compound is replaced by a *by*-phrase:

- (108) a. *Changing by weather is typical of western Massachusetts.
 b. *Elapsing by time worries philosophers.
 c. *Beating by hearts is a good sign of life.
 (109) a. ?Barking by dogs can be eliminated with a muzzle.
 b. Dancing by children is forbidden.
 c. Swimming by students is allowed only on Thursdays.

This suggests that, for those speakers who allow (107a–c), the first member of the compound does not really correspond to a subject role, but rather to a *by*-phrase, which is allowed in a compound much as a locative is. English, of course, does not allow such *by*-phrases with intransitive verbs at the sentential level, but they are typical in impersonal passive constructions in many languages (such as Dutch; see Perlmutter (1978)). These languages also typically fail to allow impersonal passives and *by*-phrases with verbs like those in (106)—so-called “unaccusative” or “ergative” verbs (Burzio (1981)). Thus, the acceptability of (107a–c) for some speakers may be related to impersonal passivization, common in compounds and nominalizations (*book buying by students*).

We now have an explanation for (106) that supports the existence of traces. Suppose the first members of these compounds are indeed external arguments.³⁸ Suppose, as suggested earlier, that they cannot be assigned a θ -role by the verb in the compound unless the affix undergoes QR. Consider now the result of applying QR to a compound like (106a) *if* the movement leaves traces:



³⁸ If these verbs are treated as “unaccusative” in English, then what I am calling “external” arguments start off as internal arguments. I appeal to Case theory to explain why they must externalize. See Williams (1981b) and the discussion of “Assume GF” in Japanese in Chomsky (1981).

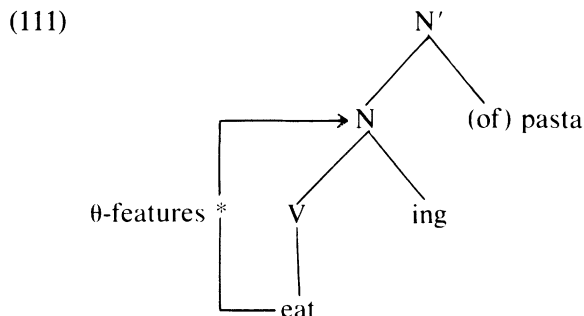
As noted earlier, traces like the sister to *change* in (110) resemble the trace of Heavy NP Shift, and such traces have the binding properties of anaphors. It is natural to treat the external argument *weather* in (110) as a subject for the purposes of the binding conditions, particularly if we accept the version of the Specified Subject Condition developed in Williams (1980; 1981b). If so, then (110) violates principle (A) of the binding theory—in particular, the Specified Subject Condition.

We can see, then, why Selkirk's Generalization holds: θ -marking of the first member of a compound forces the affix *-ing* to adjoin to the compound. If this first member is an external argument, however, this very movement, which is required for θ -marking, runs afoul of the binding conditions. Therefore, the first member cannot be an external argument. If this explanation is correct, then we have an argument for point D of the proposal with which this section began—an argument for the existence of affixal traces.

5.5. θ -Marking Again

Two problems remain unsolved. First, we have temporarily lost our explanation for Selkirk's first observation, concerning the ungrammaticality of **tree eating of pasta* (example (103b)). In explaining this observation, we took for granted the possibility of phrase-level direct θ -marking. We ruled out (103b) by arguing that *tree* was also in a θ -position and competed with the phrasal position for the direct θ -role, in violation of the θ -Criterion. Now we have argued that the first position in a deverbal compound is a θ -position only if QR applies to the deverbalizing affix. What now rules out **tree eating of pasta*?

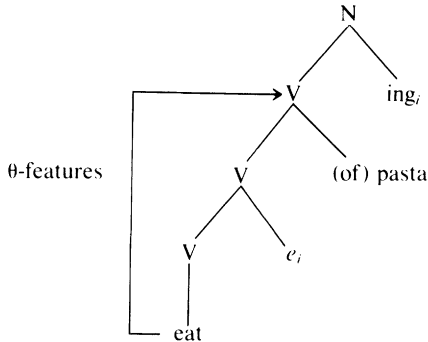
The new answer follows the same lines as the previous one, but it answers one additional question: what allows assignment of a phrasal θ -role by a deverbal noun or adjective in the first place (as in *eating of pasta*), given that the verbal θ -assigner is not the head? Lieber (1983) suggests that θ -assigning features follow the "percolation path" within a compound (cf. Walinska de Hackbeil (forthcoming)). This should mean that the presence of an affixal head like *-ing*, which percolates its own features, should prevent the percolation of a verb's θ -features to the phrasal level, ruling out the obviously good phrase *eating of pasta*:



A suggestion by Lebeaux (1984) can solve this problem and at the same time allow us

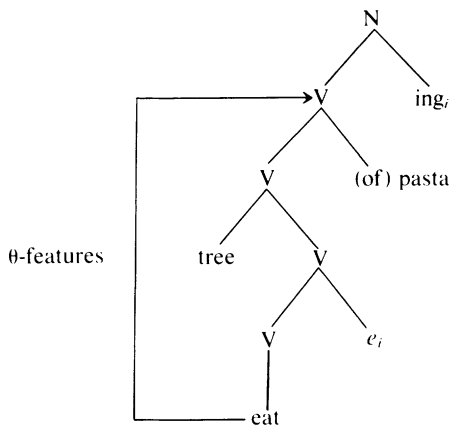
to keep our explanation of Selkirk's observation.³⁹ With reference to examples like *transformational grammarian*, I have already proposed that affix movement may sometimes take phrases, rather than words, as its domain. Lebeaux suggests that exactly such affix movement is responsible for the θ -marking of *pasta* in (111).⁴⁰ As the result of affix movement at the phrasal level, the θ -features of *eat* can percolate to the phrasal level and assign a θ -role to *pasta*:

(112)



Now suppose that *eating* has a phrasal complement *and* is part of a compound, as in (103b), **tree eating of pasta*. To assign *pasta* its θ -role, *-ing* must move, exactly as in (112). If it does not move, *pasta* will lack a θ -role, thereby violating the θ -Criterion. Moving *-ing* is necessary in order to place the phrasal complement next to a projection of V, which is the only way it can receive its θ -role. This movement, however, has other consequences. In particular, in **tree eating of pasta*, not only will *pasta* be a sister to a projection of V and receive a θ -role, but also *tree* will be a sister to a projection of V and a contender for the same θ -role, exactly as in the earlier explanation of the phenomenon:

(113)



³⁹ I ignore certain complexities of the "inheritance problem"—how derived words inherit their θ -assigning properties—discussed by Randall (1982) and Roeper (1984).

⁴⁰ The focus of Lebeaux's discussion is to account for the properties of a "verbal" reading that results from such movement from a "nominal" (often resultative) meaning that results from the absence of movement. Also, traces are not crucial to Lebeaux's discussion, as they are here.

Lebeaux's suggestion, which has many other interesting consequences for the interaction of θ -theory and morphology, thus allows us to maintain our explanation of Selkirk's observation, which was important in motivating compound-internal θ -theory in the first place.

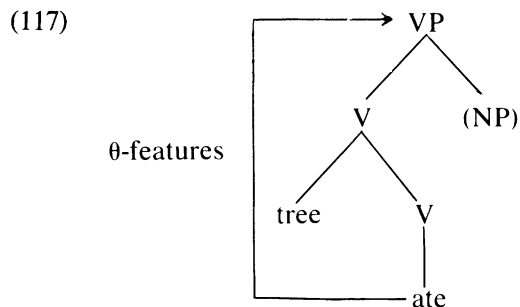
The second problem is empirical and again deals with Selkirk's observation concerning **tree eating of pasta*. J. Levi (personal communication; also see Levi (1978)) points out a number of apparent exceptions to Selkirk's observation, which are accepted by all speakers. For example:

- (114) a. home delivering of babies
 b. home canning of tomatoes
 c. lab testing of drugs
 d. spoon feeding of children
 e. hand cranking of old cars
 f. hand making of clothes

These cases are not true exceptions to the **tree eating of pasta* phenomenon, nor do they undermine the generalization in (102). Note first that cases like (114a–c) appear to be exceptions to another phenomenon as well. (115a–d) illustrate (and here I differ with some judgments in Lieber (1983)) that N–V compounds corresponding to familiar deverbal compounds are generally unacceptable; but compare the situation in (116a–f):

- (115) a. *Bill tree-ate (his pasta).
 b. *Mary drug-took (morphine).
 c. *Harry likes to river-swim (the Mississippi).
 d. *Sue book-read (*War and Peace*).
 (116) a. We home-delivered our babies.
 b. We home-can all our tomatoes.
 c. We lab-tested these drugs.
 d. We spoon-fed Junior.
 e. We hand-cranked the old Model T.
 f. We hand-made all our clothes.

To explain (115a–d), I note with Lieber that when the head and the θ -assigner in a compound are the same, the θ -assigning features (henceforth θ -features) of the head should percolate to the phrasal node:



In such a case, Lieber assumes that an unnamed principle *requires* the θ -role to be assigned at the phrase level, and not internal to the compound. I believe the appropriate principle is subcategorization, as discussed by Chomsky (1981). The role of subcategorization in that theory is to require a θ -position to exist whenever the θ -role involved is direct: hence, there are unassigned indirect θ -roles, as in action nominals with no subject (again see Lebeaux (1984)), but no unassigned direct θ -roles. The situation in (117) is now no different from the situation in (113). Not only is there a (here) obligatory phrasal θ -position that must receive the verb's direct θ -role, but there is as well a compound-internal θ -position competing for the same θ -role. The result violates the θ -Criterion, and (115a–d) are correctly ruled out.

It is encouraging, then, that within our theory we account for the examples in (115) and for **tree eating of pasta* in the same way, and that *home + deliver* and the other examples are uniformly exceptional with respect to both phenomena. But why should they be exceptional? I suggest that *home*, etc., are not arguments and do not contend for θ -roles at all. Rather, they have been reanalyzed as prefixes, and the structures that contain them are not really compounds, but normal prefixed verbs and deverbal words. It is therefore expected that all the stars derived from the θ -Criterion will not apply to them.

First note that the phenomenon identified by Levi is sporadic and semiproductive at best:⁴¹

- (118) a. **We house-delivered our babies./*house delivery of babies*
- b. **We shed-canned our tomatoes./*shed-canning of tomatoes*
- c. **We chopstick-fed Junior./*chopstick-feeding of babies*
- d. **The contortionist foot-made his clothes./*foot-making of clothes*

Most interestingly, Levi's examples differ from normal compounds in one important way. In normal deverbal compounds, the first member may be not only a noun, but also a phrase, as long as it contains neither complements nor determiners:

- (119) a. The fetishist was a gloved-hand painter.
- b. John is a silver-spoon collector.
- c. Private-home construction was forbidden by the government.

Examples of Levi's type never have a phrase as first member:

- (120) a. **It was so cold that I had to gloved-hand crank the Model T.*
- b. **The winners of the lottery can now silver-spoon feed their kids.*
- c. **I disapprove of private-home canning of tomatoes.*

These facts support the contention that the first member of Levi's examples is not a nominal argument, subject to the θ -Criterion, but has been reanalyzed as a prefix. Our

⁴¹ Also note lexical drift:

- (i) The contortionist hand-made a sweater using only his feet.
- (ii) The restaurant's pies are home-made in their kitchen.

θ -theory therefore stands. With it stands our explanation for Selkirk's Generalization. And with that explanation we have evidence for affixal traces, which is what we set out to find.

6. Conclusions: Is There a Lexicon?

I have argued that "bracketing paradoxes" actually involve representations of words at two levels, S-structure and LF, and that the mapping between these levels is an adjunction rule called "QR," which leaves traces. As noted, my more specific proposals are contingent on the more general ones, but not vice versa. In any case, only if representations like the proposed LF representations are "real" and have well-defined properties can we account in a natural way for all the phenomena discussed—as the result of the interaction of lexical properties with the general structure of the grammar.

Although I have argued that a number of well-known syntactic principles apply word-internally as well as phrasally, there is nothing a priori true about such claims, nor are they a null hypothesis. To assume a priori that principles below the word level are the same as those above the word level makes as much sense as an a priori belief that metrical structure or the feature [+continuant] plays a role in the formulation of *Wh* Movement. Nonetheless, one theoretical tack that has proven extremely fruitful both in syntax and in phonology may be incorporated with profit in morphology. This is the notion of "level of representation" and the attendant modular view of grammatical structure. This has been the main point of our discussion.

Not surprisingly, once we begin to think of morphology in the modular way suggested here, questions arise concerning what the modules are. I want to end the discussion with a question that stems from the work presented here. If we grant that word structure may be distinct at LF and at S-structure, is there a component called the lexicon?

We discussed in section 3 some "rules of idiosyncratic interpretation" that allow combinations of morphemes to have a special interpretation. We argued that these rules require a form of sisterhood at LF, where they apply. Perhaps *all* rules governing the idiosyncratic properties of lexical items are subject to a sisterhood or government requirement.

Now it is rules of exactly this type that provide the best argument for a lexical component with interesting properties. It is often claimed that a lexical component is precisely a storehouse of the idiosyncratic properties of lexical items, perhaps with certain "lexical rules" eliminating redundancy in this storehouse. Our discussion makes it necessary to reexamine whether such a storehouse, containing facts about *lexical items*, exists.

Consider the word *unruly*. Given the nonexistence of **ruly* (for most speakers), *unruly* must get its meaning ('ill-behaved', 'unorderly') by virtue of a rule of idiosyncratic interpretation. Now consider the comparative form *unrulier*. This form obviously undergoes the same rule of idiosyncratic interpretation as does the positive form *unruly*. We

have argued for a sisterhood requirement on such rules and for strict binary branching in word structure. This means that the rule of idiosyncratic interpretation here must apply to *un-* and *ruly* in the following structure (ignoring traces):

(121) [[un + ruly] er]

But now recall the phonological condition governing comparative *-er*. To solve the problem described in section 2, this condition must also be subject to a sisterhood requirement. Comparative *-er* may be the sister of a monosyllable or a near disyllable like *happy*, but not the sister of a polysyllable like *unruly* or *unhappy*. The condition on *-er* thus motivates the structure in (122):

(122) [un [ruly + er]]

Clearly, the grammar has one rule governing the idiosyncratic interpretation of *un-* and *ruly* and another rule or condition governing the attachment of *-er* to *ruly*, but each rule applies at a different level. Also, there is no rule that tells us all we need to know about *unrulier* as a whole. Should we wish, we could obviously define a "lexicon" as the set of all such idiosyncratic rules, with a legend dictating which type of rule applies at which level, but does it make any sense to call this set a component? Does it have any well-defined properties besides the sisterhood requirement, which characterizes many other aspects of grammar?

If there is no lexicon, an alternative view of "lexical phenomena" might be constructed as follows: a language learner acquires morpheme-specific rules of differing types, which apply at different levels of representation. Ideally, the type of rule will dictate the level at which it applies, but there may be cases of the sort discussed by Borer (1984a,b) in which one and the same rule applies at different levels in different languages. These rules involve sets of nodes no bigger than pairs, the normal association of individual morphemes with individual meanings and sounds being the limiting case. Call these rules a "lexicon" if you will, but it is not necessarily a coherent component with properties specific to the component as a whole; rather, the properties of the idiosyncratic rules are dictated by the general properties of the levels at which they apply. It would be perhaps as profitable to refer to a "lexical component" as it is to refer to a "Move α component," when Move α is a rule whose properties at each level are determined by that level.

Be that as it may, I do believe that importing the notion of "level of representation" into morphology is profitable, and at least makes previously somewhat odd and mysterious "bracketing paradoxes" turn out to be less mysterious and not at all paradoxical.

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